

Config 10 (0.05|0.9|1|1|0.52|0.5| 0.7| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.740 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9012

FWER (P reject ≥ 1 | H0) : 0.0517

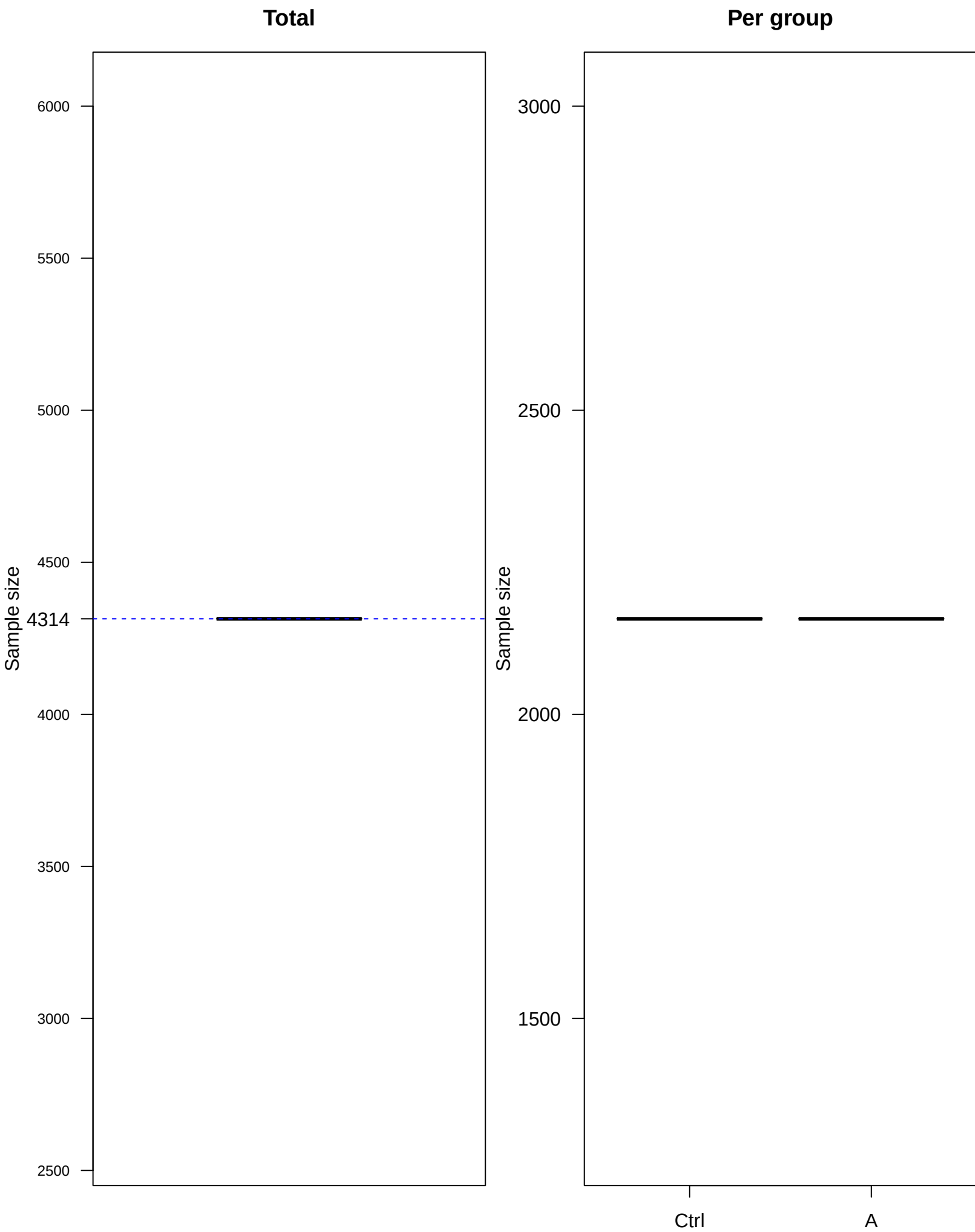
Max sample size (planned) : 4314

ESS under H1 (overstated) : 4314.0

ESS under H0 (overstated) : 4314.0

NB: design powered for the binary test (δ/σ), so achieved power should track the target; residual gap is small-sample discreteness. fut.arm.nostop disables futility stopping, so ESS is overstated – see ../binary-binding for realistic ESS.

Config 10 – sample size (H1) [binary]



Config 12 (0.05|0.9|1|1|0.52|0.5| 0.9| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.900

effect pi_t = 0.940 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9048

FWER (P reject ≥ 1 | H0) : 0.0489

Max sample size (planned) : 1572

ESS under H1 (overstated) : 1572.0

ESS under H0 (overstated) : 1572.0

NB: design powered for the binary test (delta/sd), so achieved

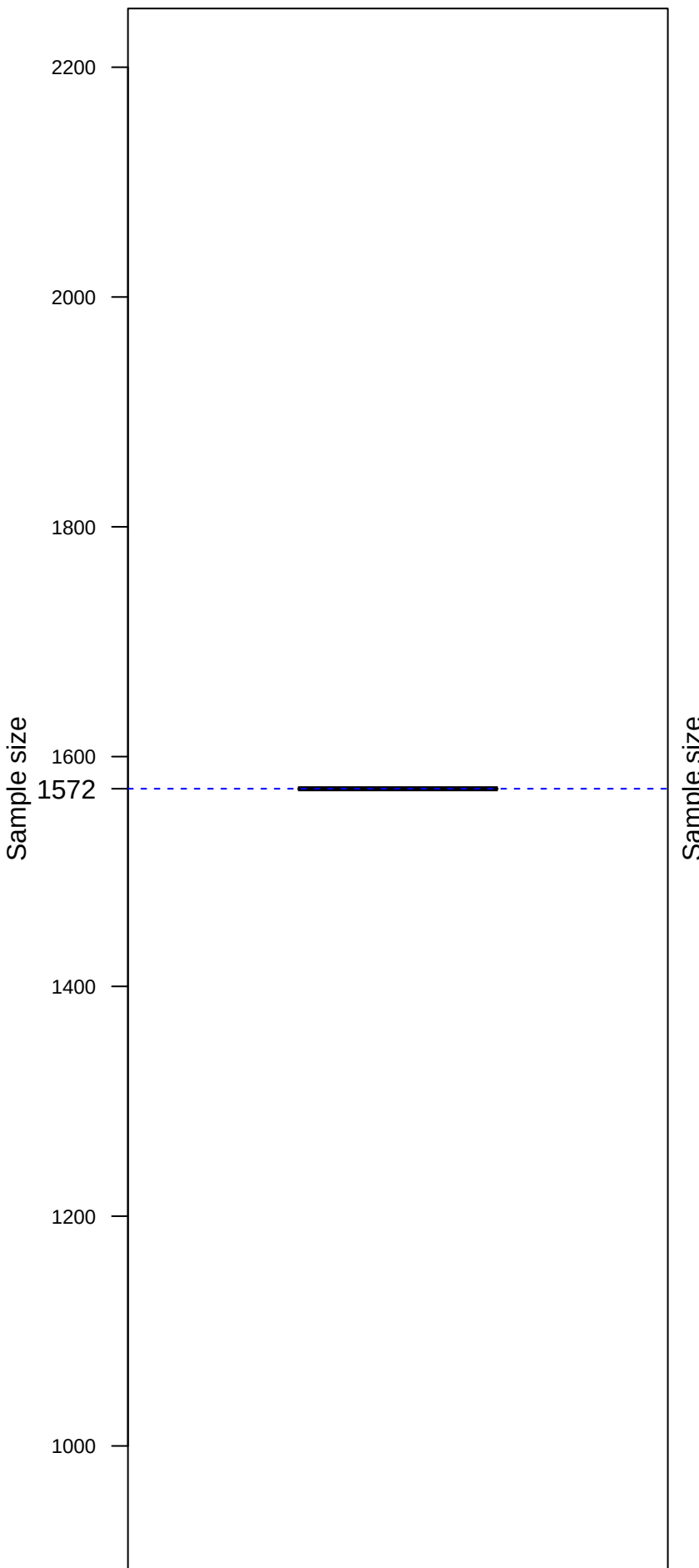
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

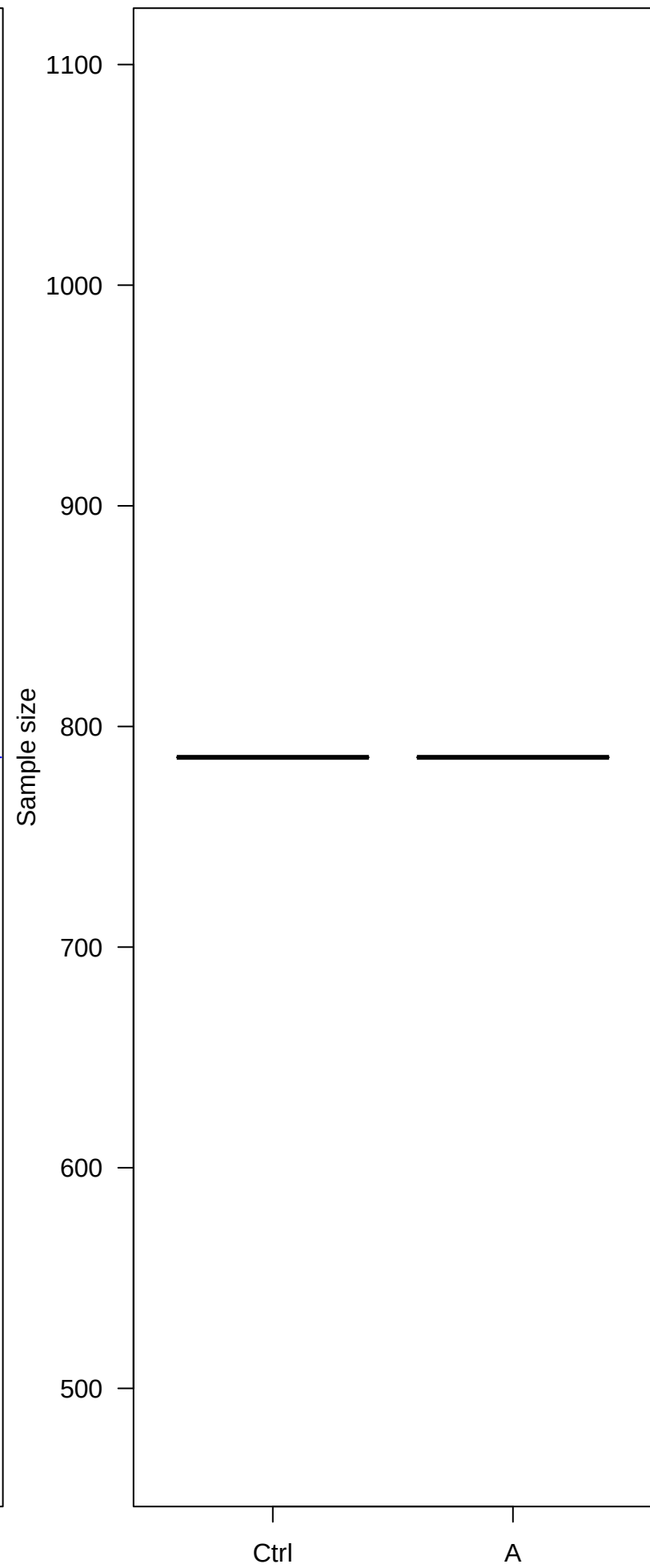
is overstated – see ../binary-binding for realistic ESS.

Config 12 – sample size (H1) [binary]

Total



Per group



Config 20 (0.05|0.9|1|1|0.55|0.5| 0.5| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.500$

effect $\pi_t = 0.600$ ($= \pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9023

FWER (P reject ≥ 1 | H0) : 0.0532

Max sample size (planned) : 846

ESS under H1 (overstated) : 846.0

ESS under H0 (overstated) : 846.0

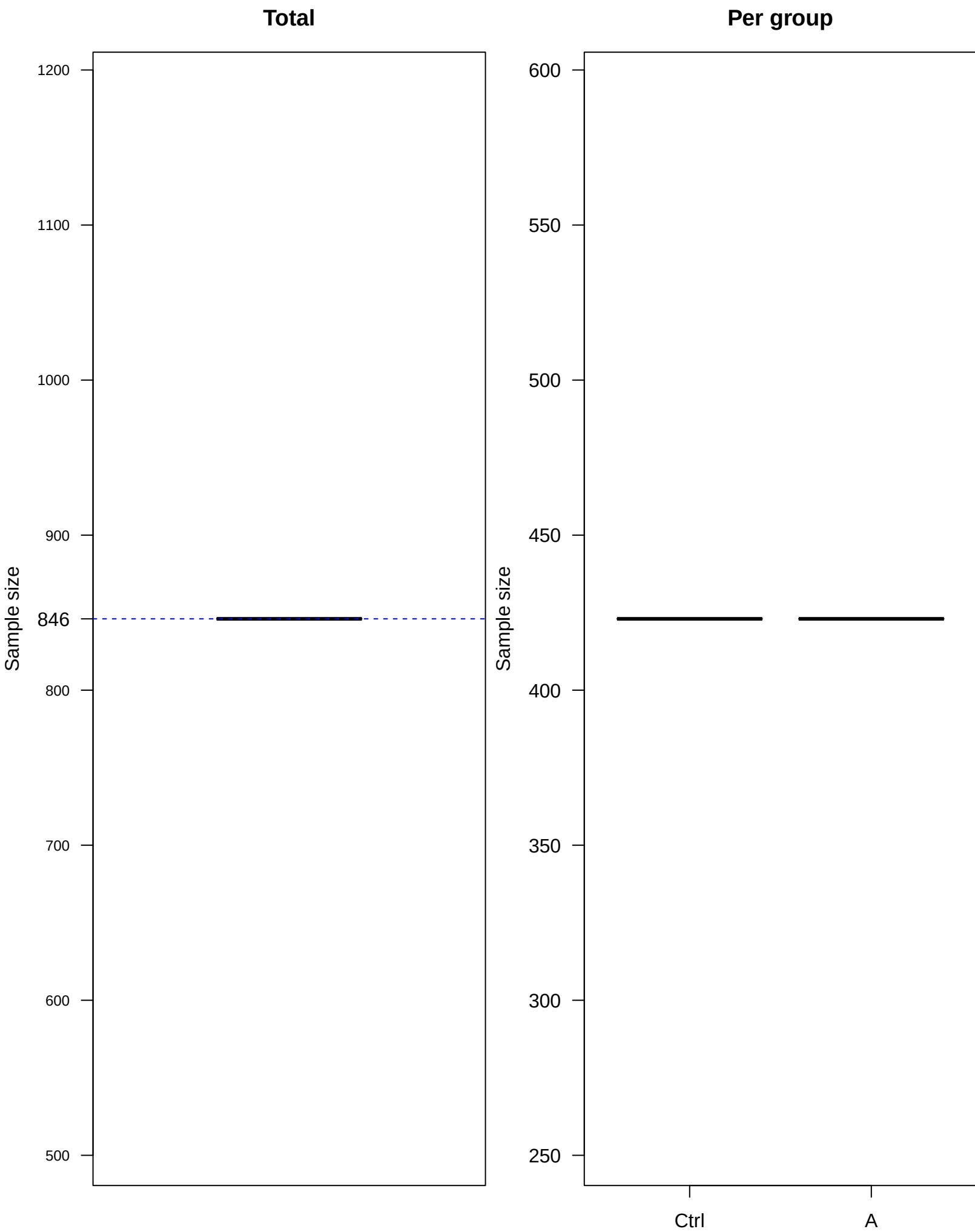
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 20 – sample size (H1) [binary]



Config 34 (0.05|0.9|1|1| 0.6|0.5| 0.7| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.900 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9126

FWER (P reject ≥ 1 | H0) : 0.0524

Max sample size (planned) : 134

ESS under H1 (overstated) : 134.0

ESS under H0 (overstated) : 134.0

NB: design powered for the binary test (delta/sd), so achieved

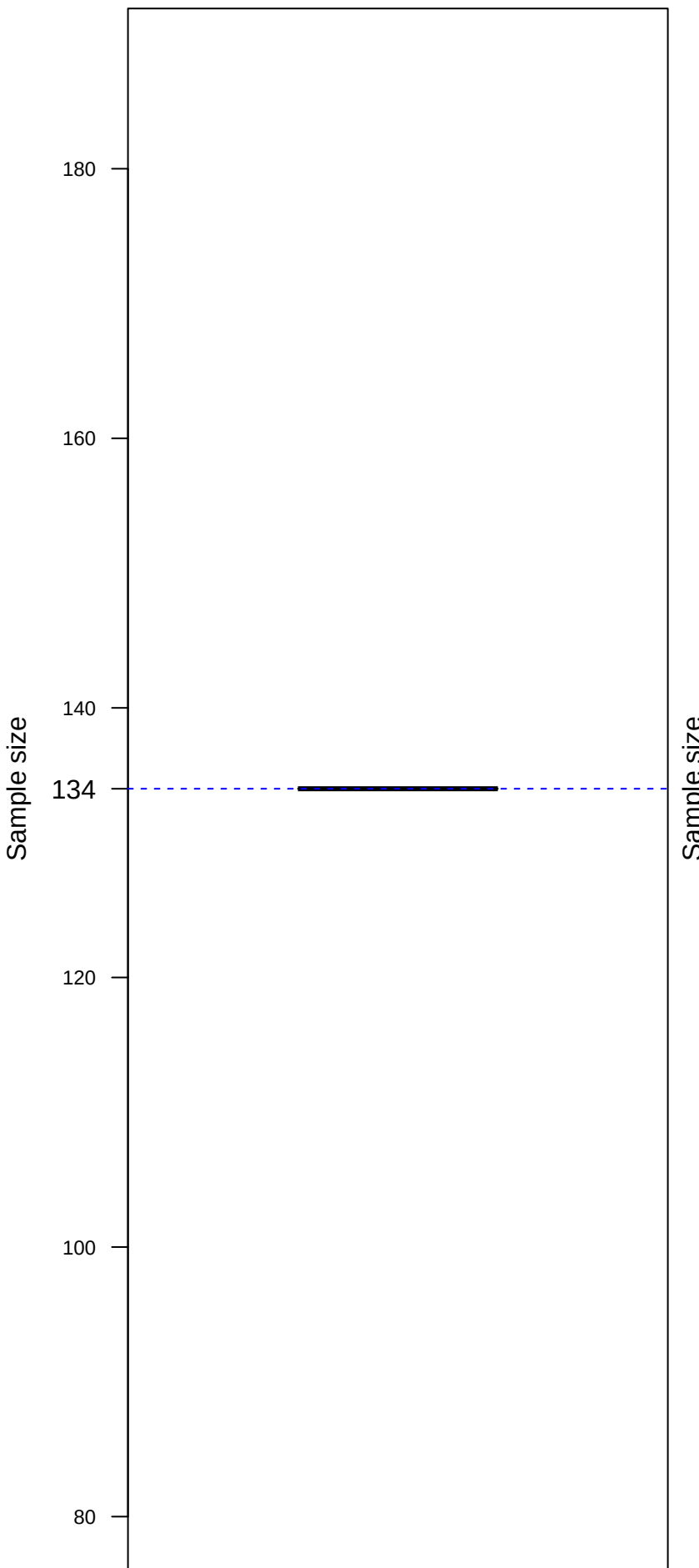
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

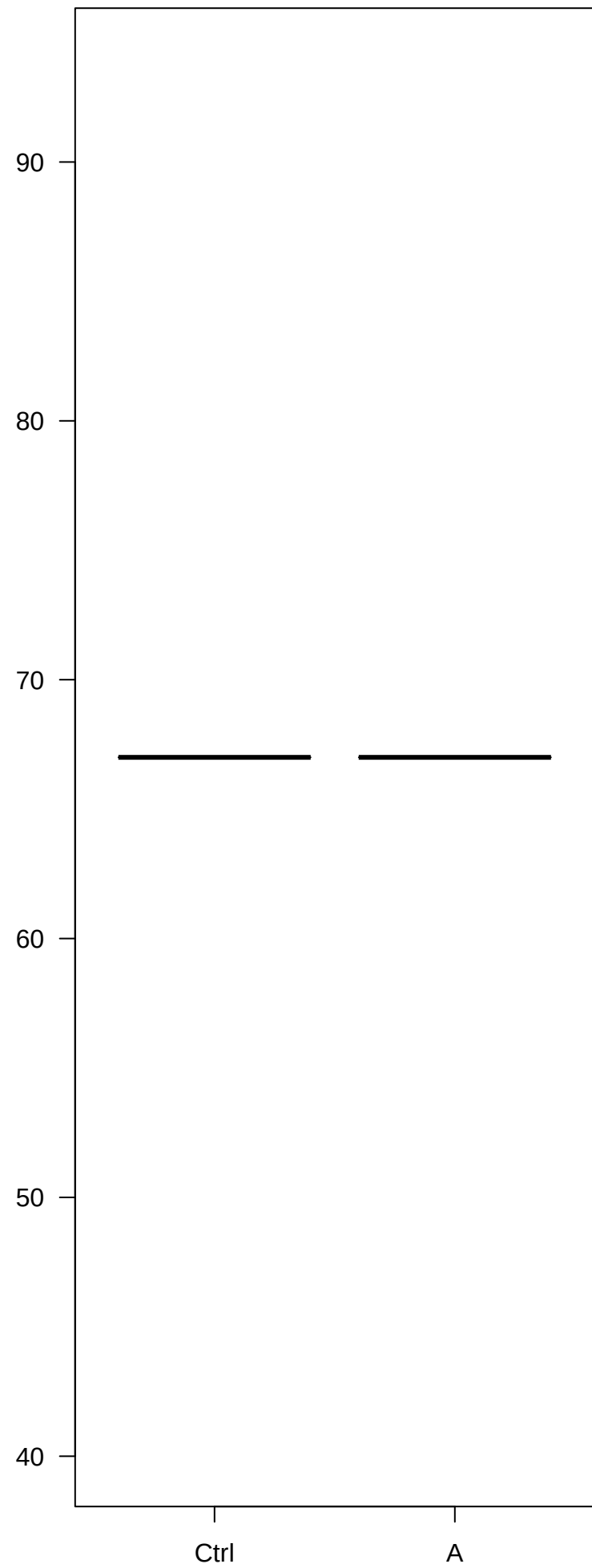
is overstated – see ../binary-binding for realistic ESS.

Config 34 – sample size (H1) [binary]

Total



Per group



Config 38 (0.05|0.9|1|1|0.65|0.5|0.05| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.350$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9314

FWER (P reject ≥ 1 | H0) : 0.0450

Max sample size (planned) : 58

ESS under H1 (overstated) : 58.0

ESS under H0 (overstated) : 58.0

NB: design powered for the binary test (δ/σ), so achieved

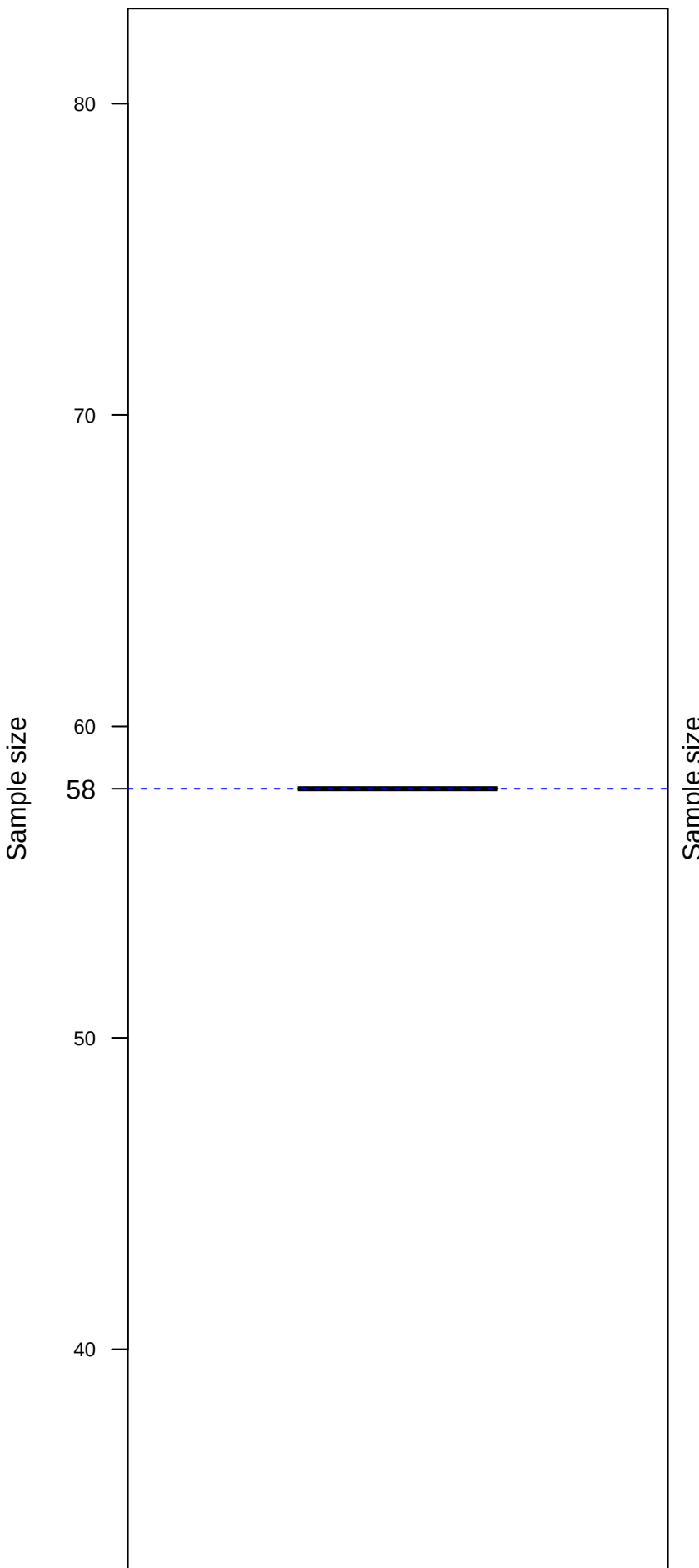
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

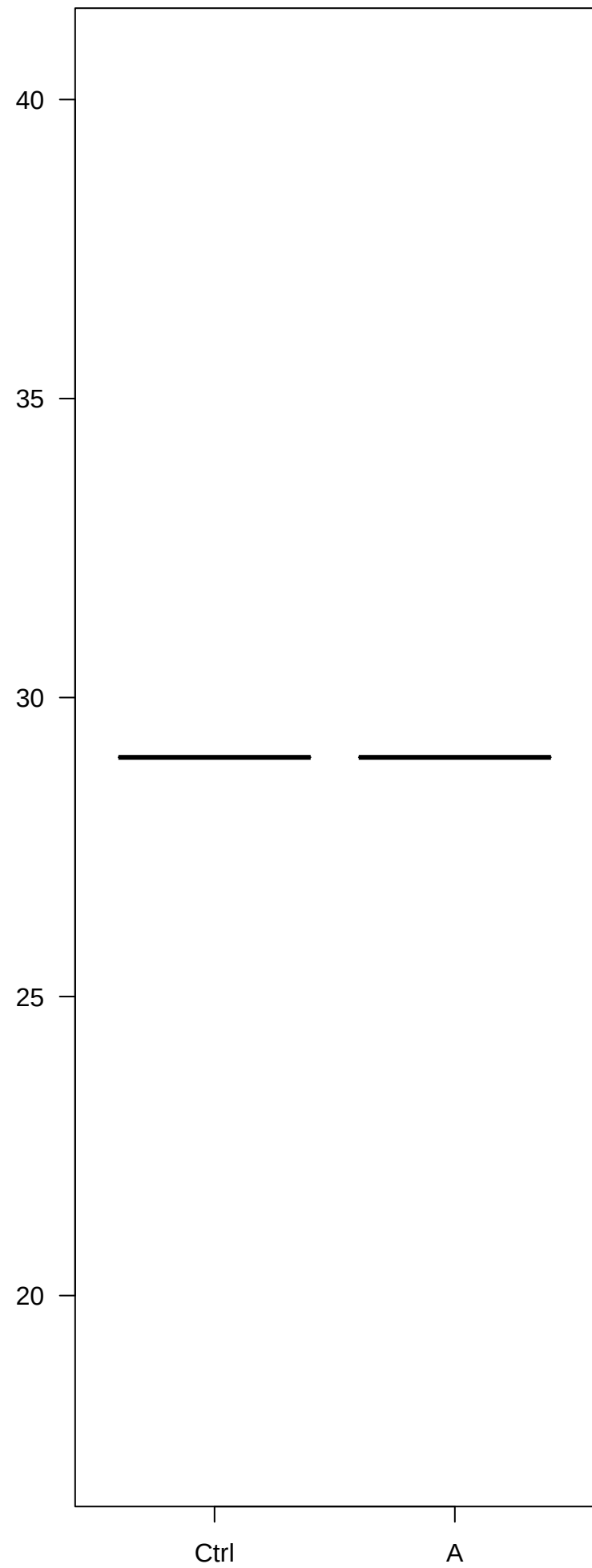
is overstated – see ../binary-binding for realistic ESS.

Config 38 – sample size (H1) [binary]

Total



Per group



Config 42 (0.05|0.9|1|1|0.65|0.5| 0.3| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9157

FWER (P reject ≥ 1 | H0) : 0.0508

Max sample size (planned) : 92

ESS under H1 (overstated) : 92.0

ESS under H0 (overstated) : 92.0

NB: design powered for the binary test (delta/sd), so achieved

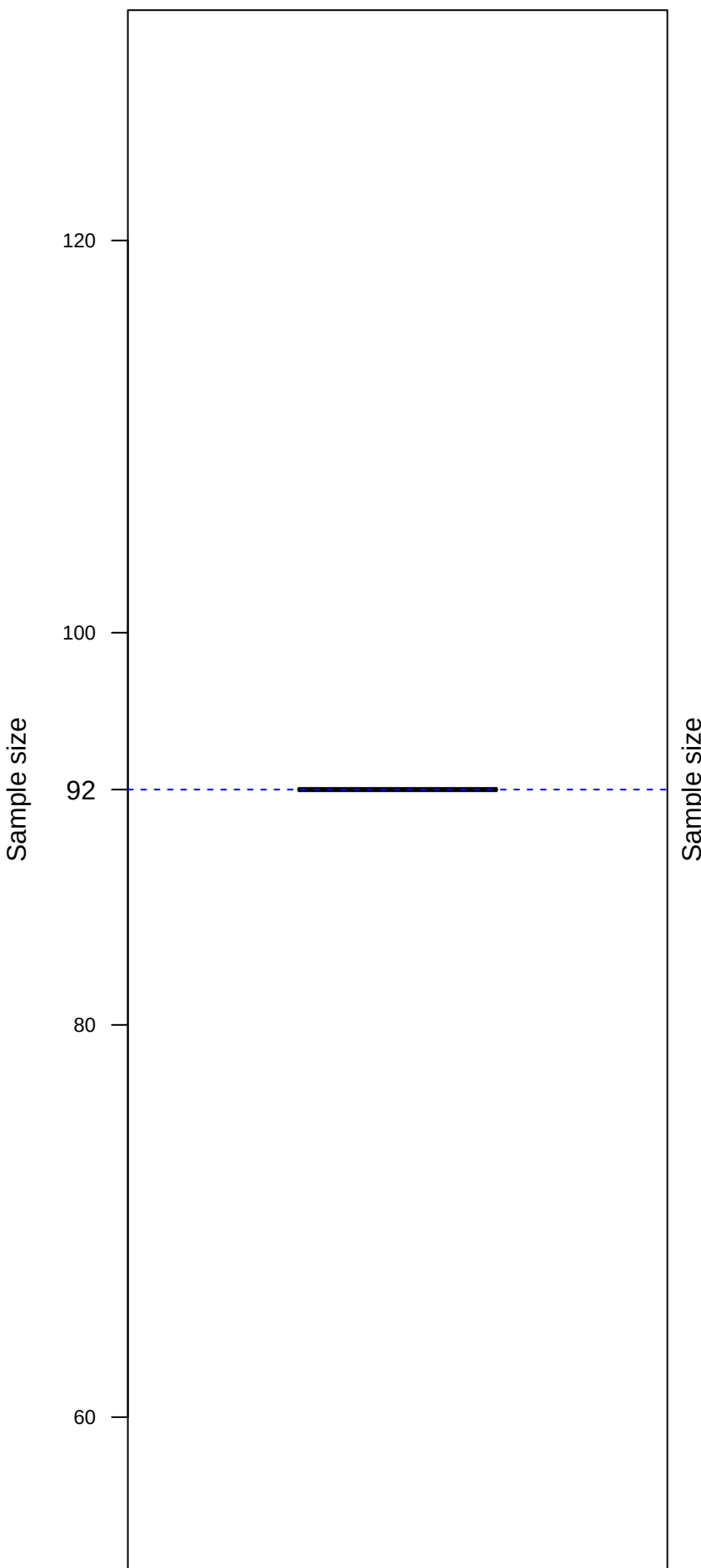
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

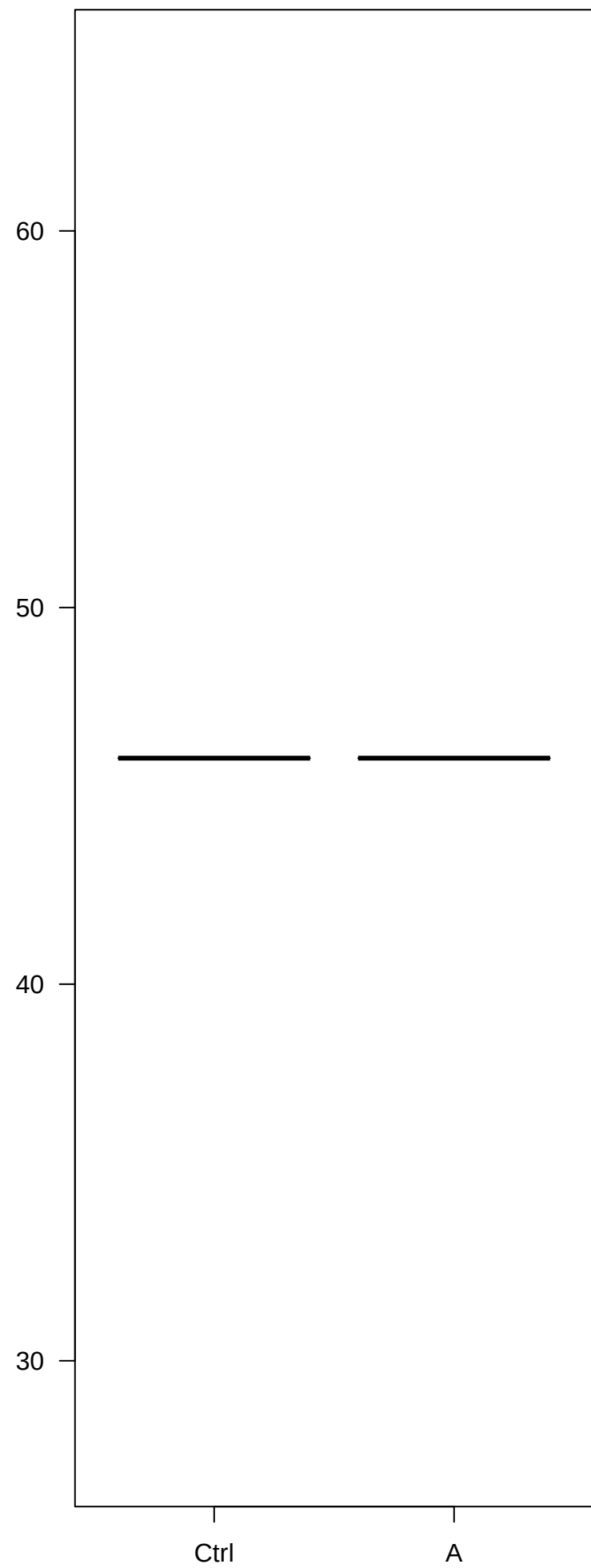
is overstated – see ../binary-binding for realistic ESS.

Config 42 – sample size (H1) [binary]

Total



Per group



Config 44 (0.05|0.9|1|1|0.65|0.5| 0.5| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9026

FWER (P reject ≥ 1 | H0) : 0.0544

Max sample size (planned) : 84

ESS under H1 (overstated) : 84.0

ESS under H0 (overstated) : 84.0

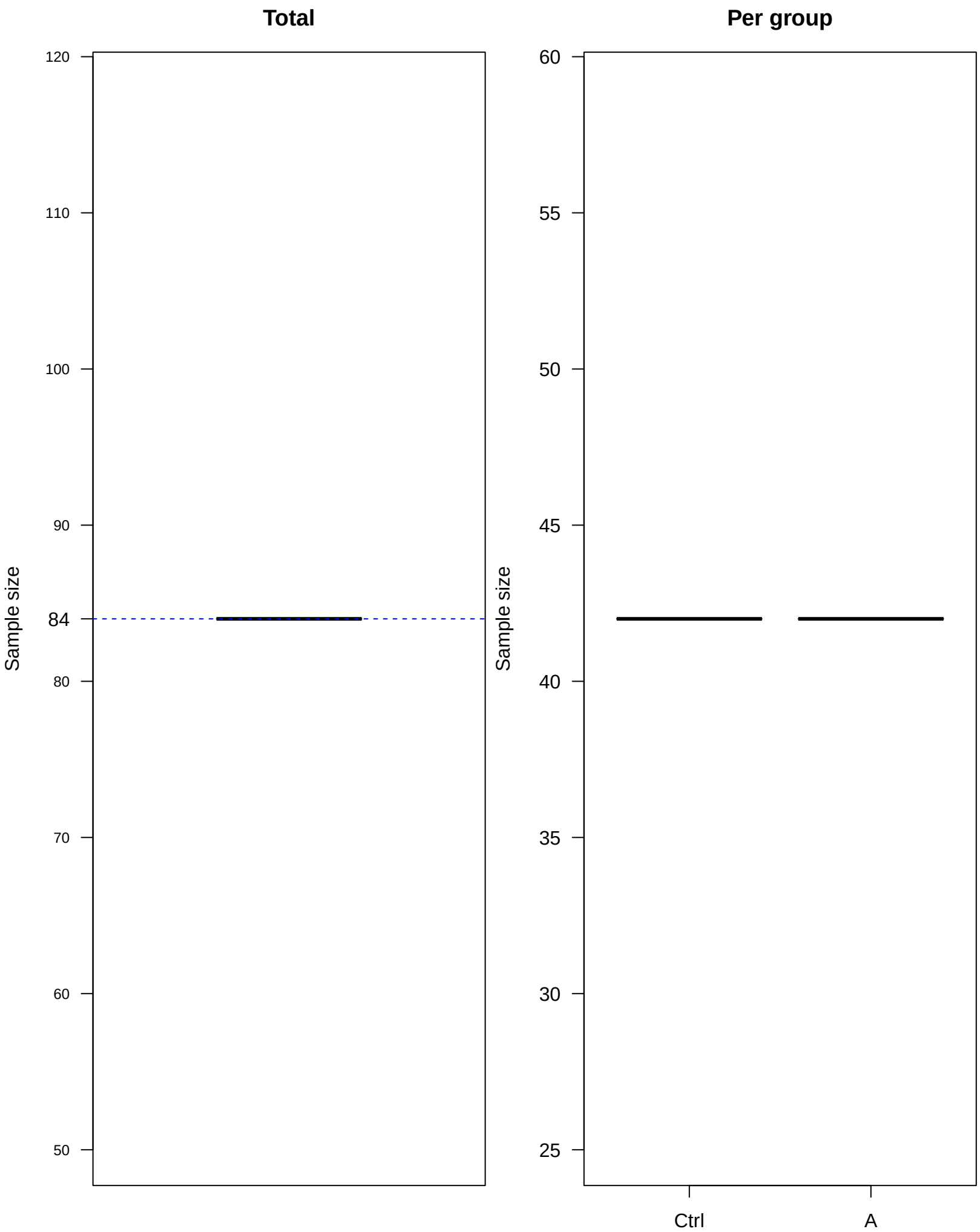
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 44 – sample size (H1) [binary]



Config 54 (0.05|0.9|1|1|0.75|0.5| 0.3| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9195

FWER (P reject ≥ 1 | H0) : 0.0473

Max sample size (planned) : 32

ESS under H1 (overstated) : 32.0

ESS under H0 (overstated) : 32.0

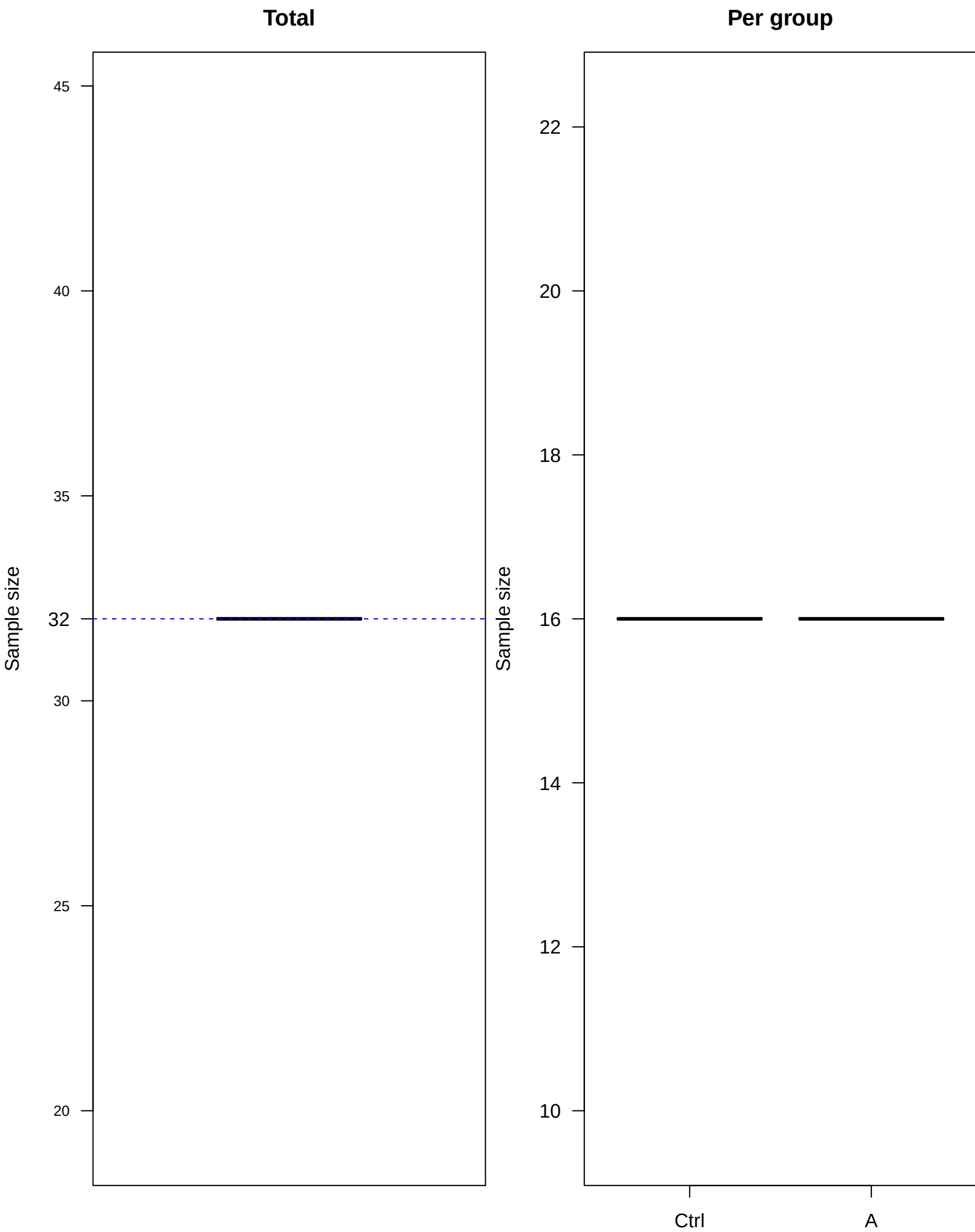
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 54 – sample size (H1) [binary]



Config 64 (0.05|0.9|1|1|0.85|0.5| 0.1| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.100

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9230

FWER (P reject ≥ 1 | H0) : 0.0134

Max sample size (planned) : 14

ESS under H1 (overstated) : 14.0

ESS under H0 (overstated) : 14.0

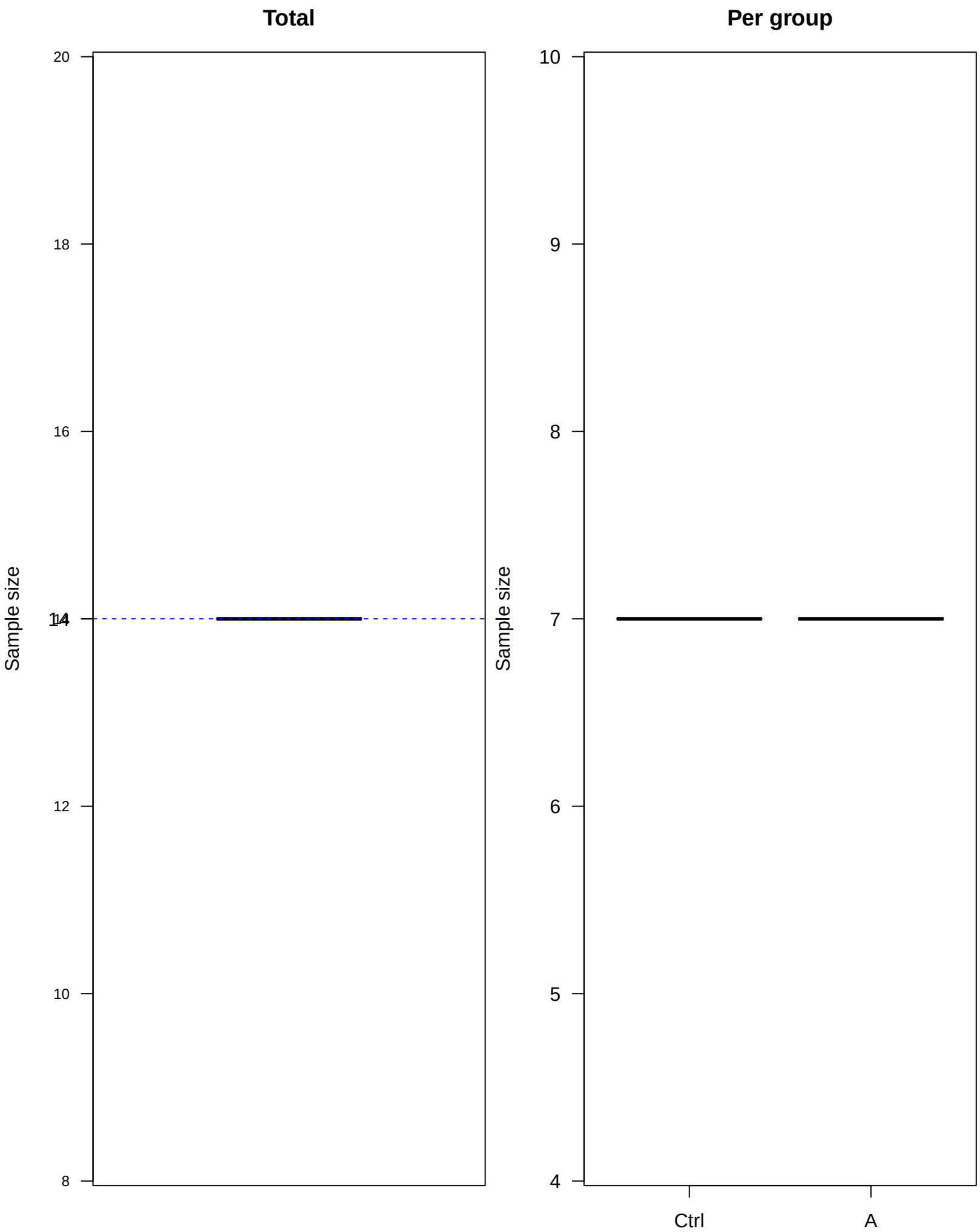
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 64 – sample size (H1) [binary]



Config 74 (0.05|0.9|1|1|0.95|0.5|0.05| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.950$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9673

FWER (P reject ≥ 1 | H0) : 0.0064

Max sample size (planned) : 6

ESS under H1 (overstated) : 6.0

ESS under H0 (overstated) : 6.0

NB: design powered for the binary test (δ/σ), so achieved

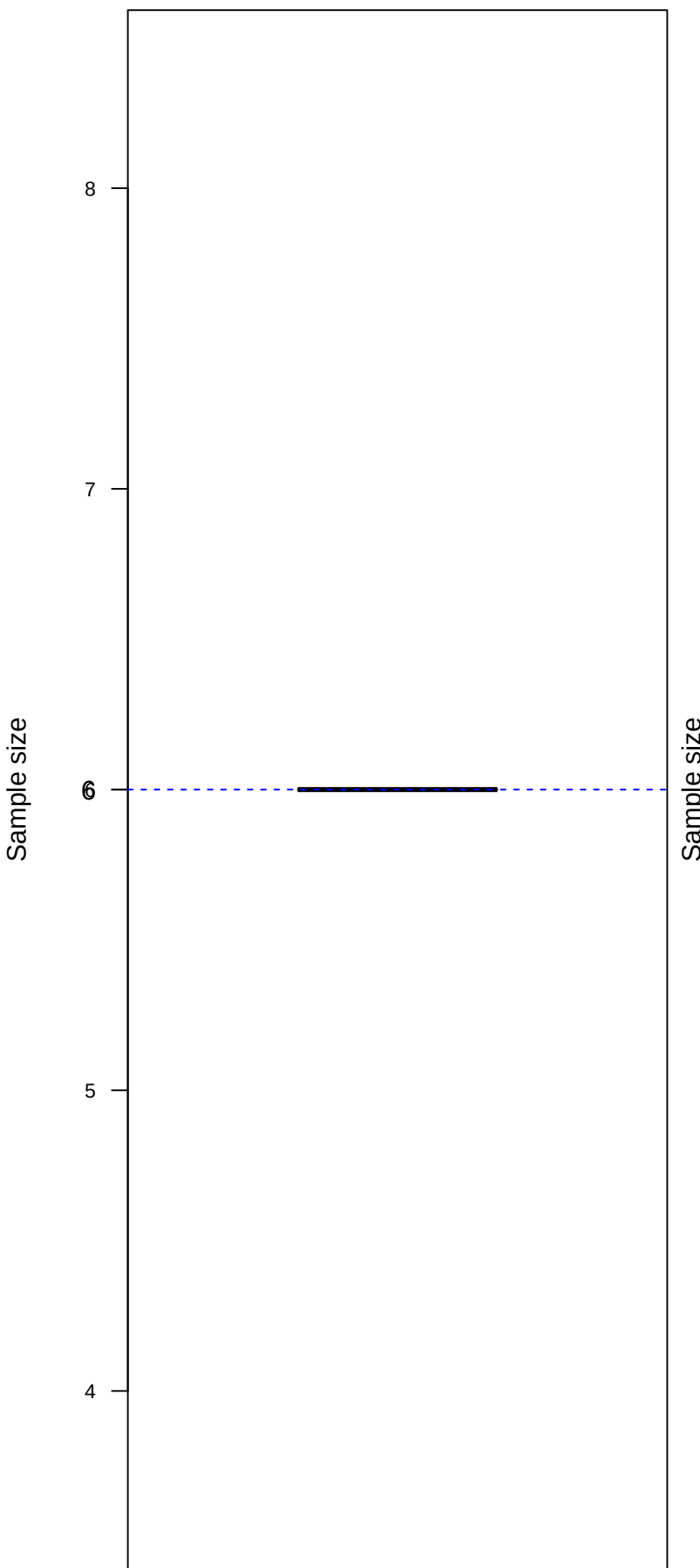
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

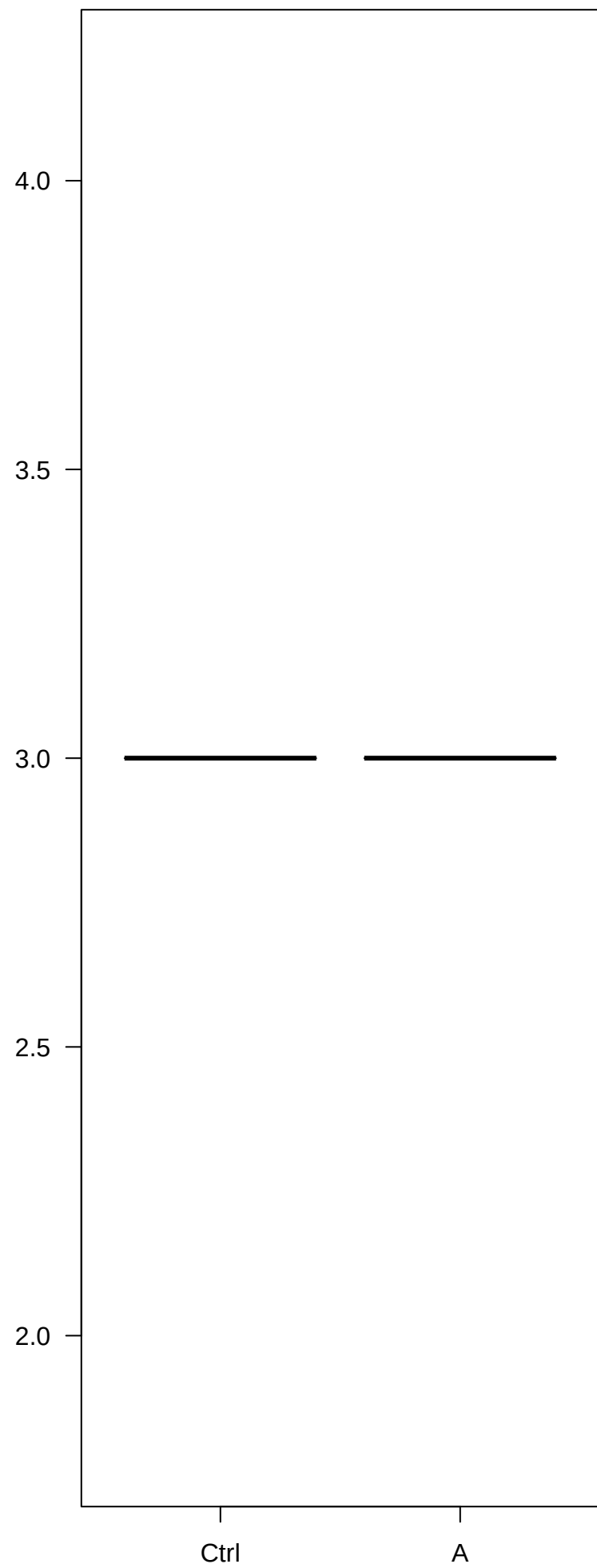
is overstated – see ../binary-binding for realistic ESS.

Config 74 – sample size (H1) [binary]

Total



Per group



Config 94 (0.05|0.9|1|2|0.52|0.5| 0.7| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.740 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9290

FWER (P reject ≥ 1 | H0) : 0.0512

Max sample size (planned) : 4904

ESS under H1 (overstated) : 3841.5

ESS under H0 (overstated) : 4879.3

NB: design powered for the binary test (delta/sd), so achieved

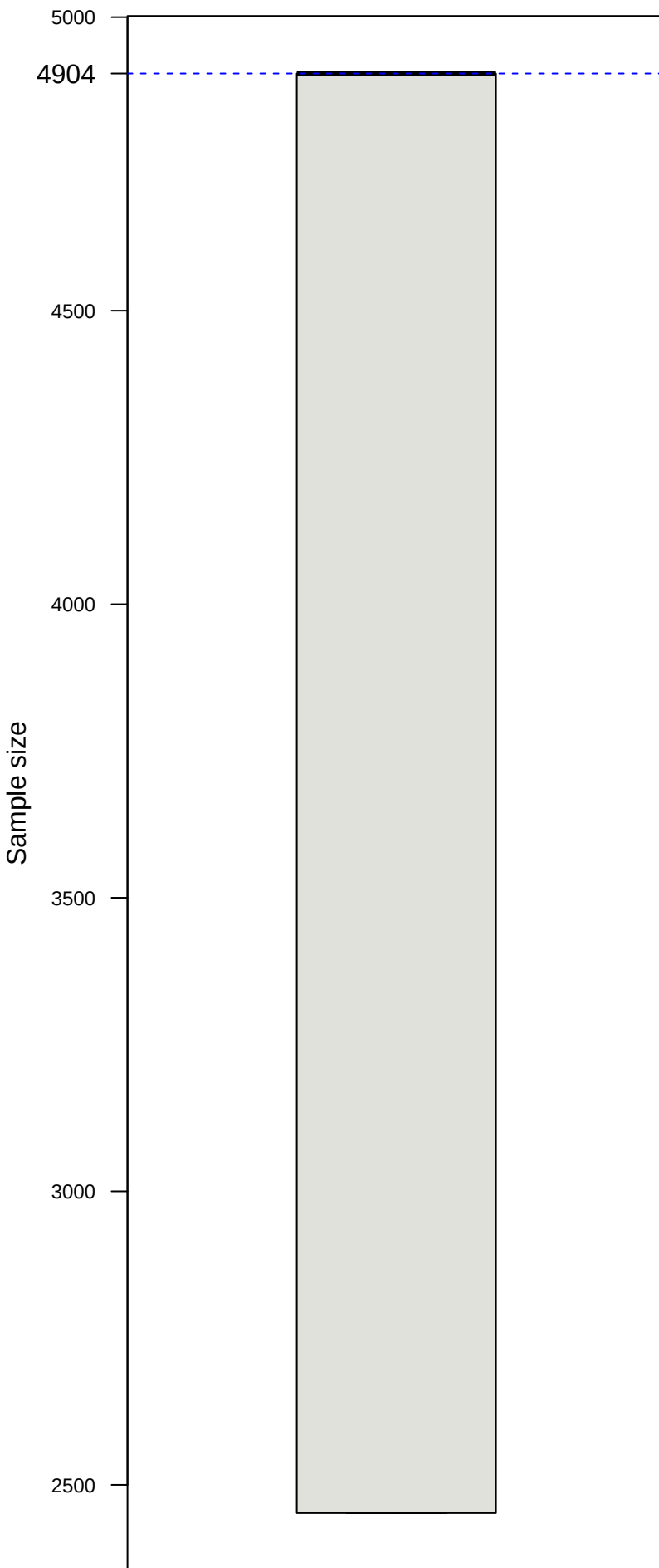
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

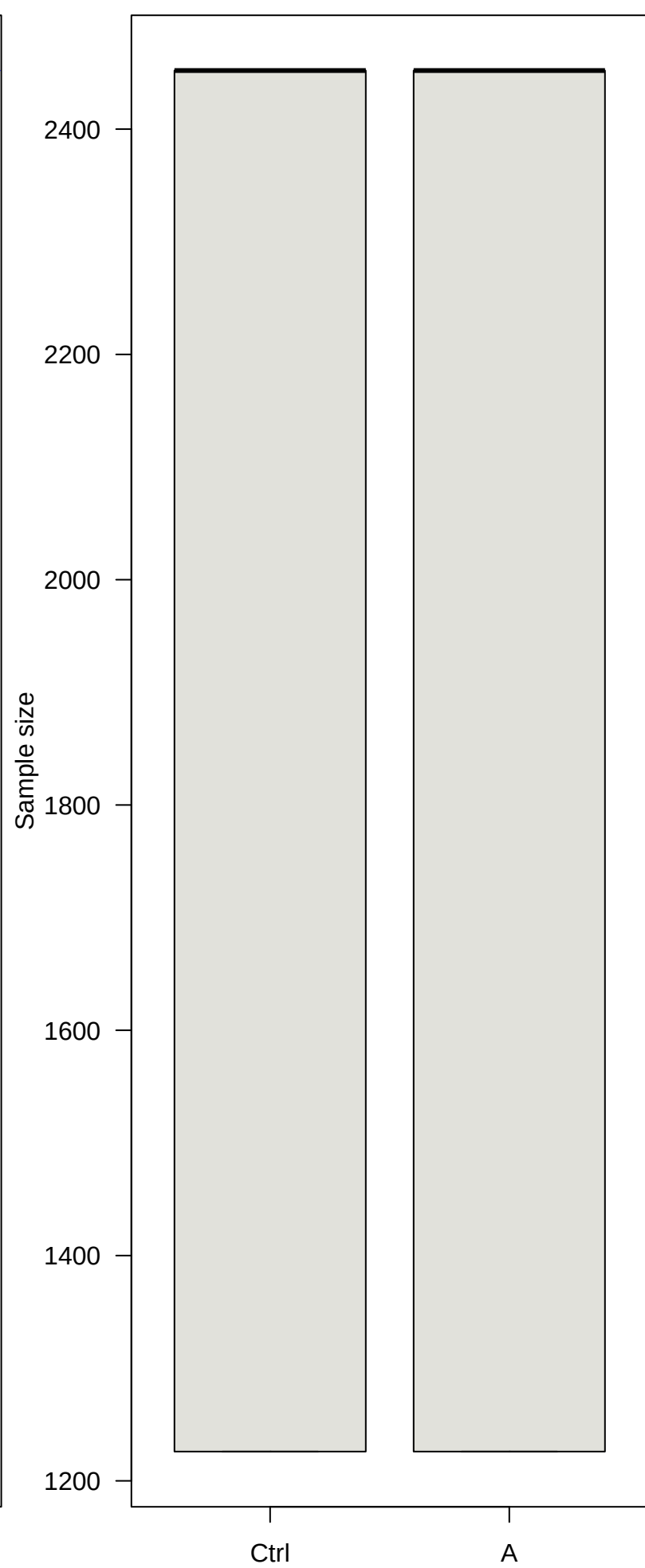
is overstated – see ../binary-binding for realistic ESS.

Config 94 – sample size (H1) [binary]

Total



Per group



Config 96 (0.05|0.9|1|2|0.52|0.5| 0.9| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.900

effect pi_t = 0.940 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9278

FWER (P reject ≥ 1 | H0) : 0.0493

Max sample size (planned) : 1788

ESS under H1 (overstated) : 1404.7

ESS under H0 (overstated) : 1779.7

NB: design powered for the binary test (delta/sd), so achieved

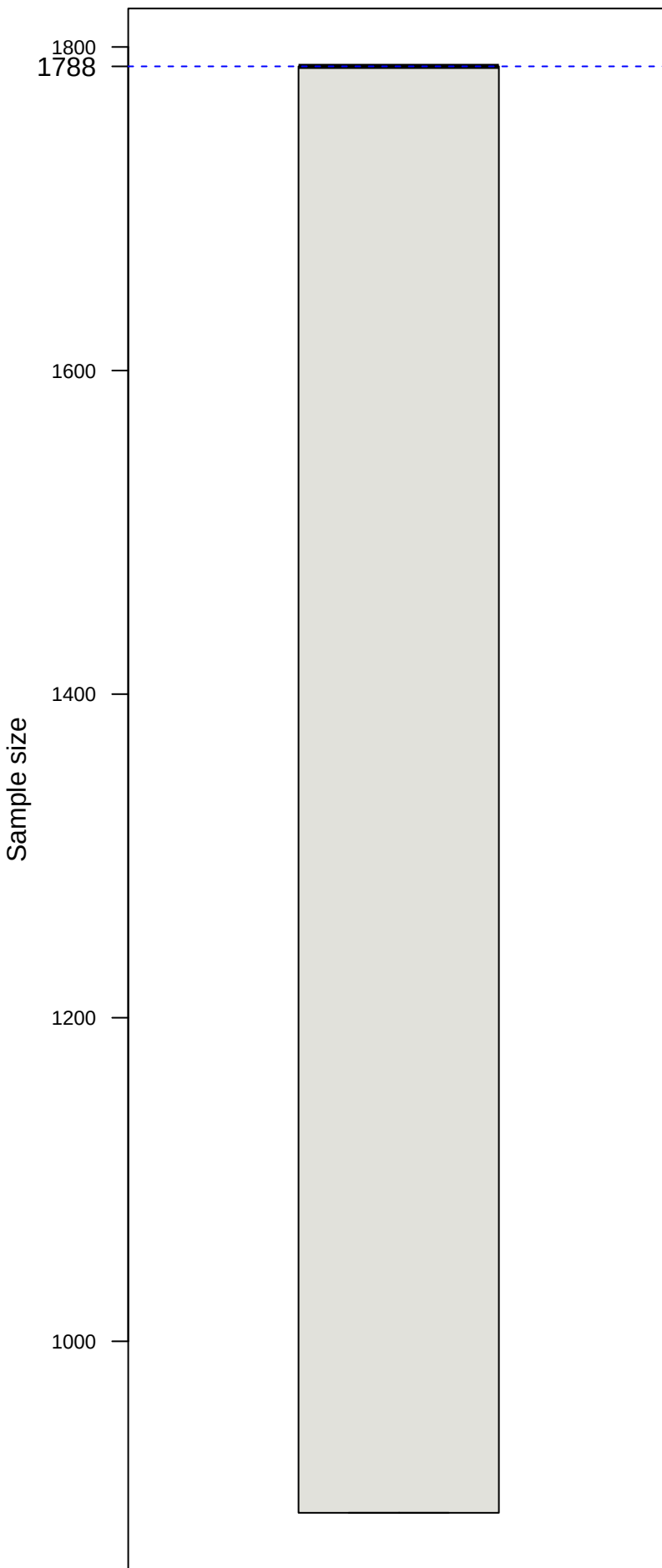
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

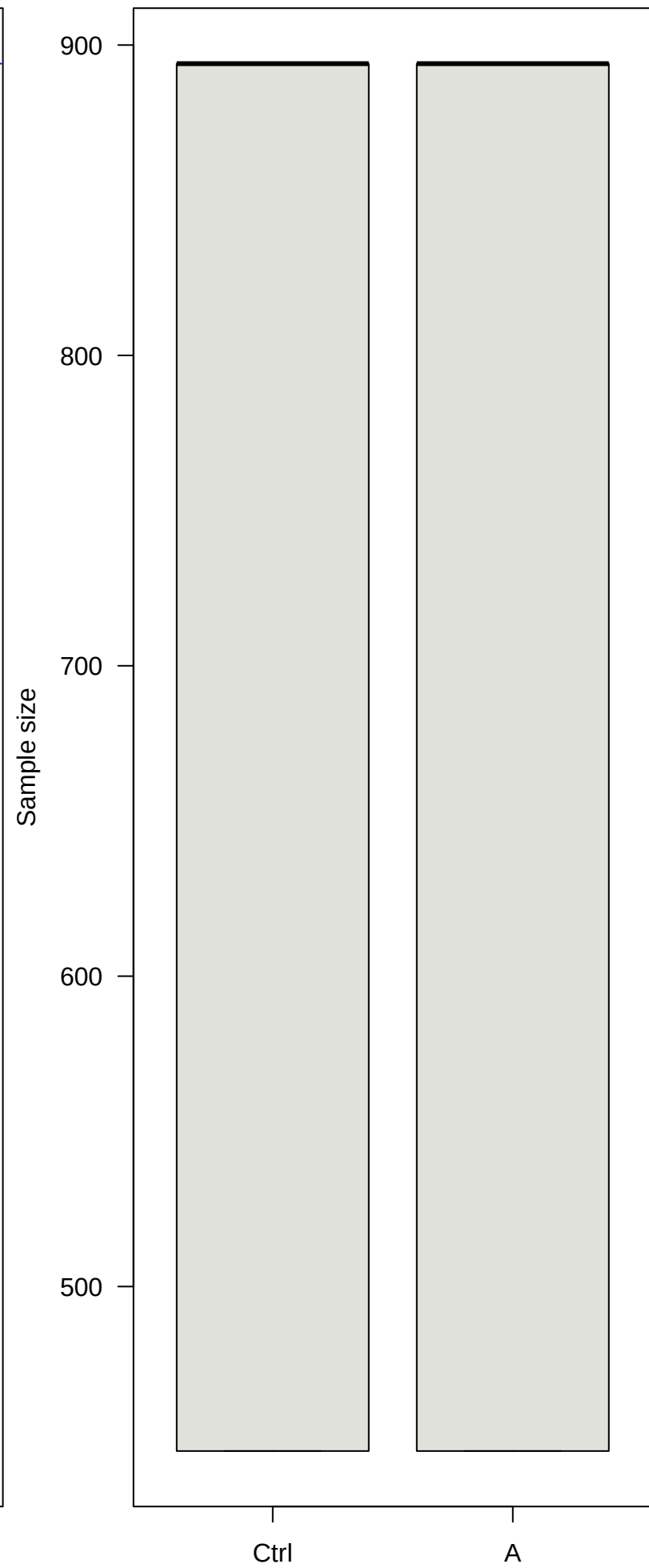
is overstated – see ../binary-binding for realistic ESS.

Config 96 – sample size (H1) [binary]

Total



Per group



Config 104 (0.05|0.9|1|2|0.55|0.5| 0.5| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9298

FWER (P reject ≥ 1 | H0) : 0.0545

Max sample size (planned) : 960

ESS under H1 (overstated) : 749.1

ESS under H0 (overstated) : 955.1

NB: design powered for the binary test (delta/sd), so achieved

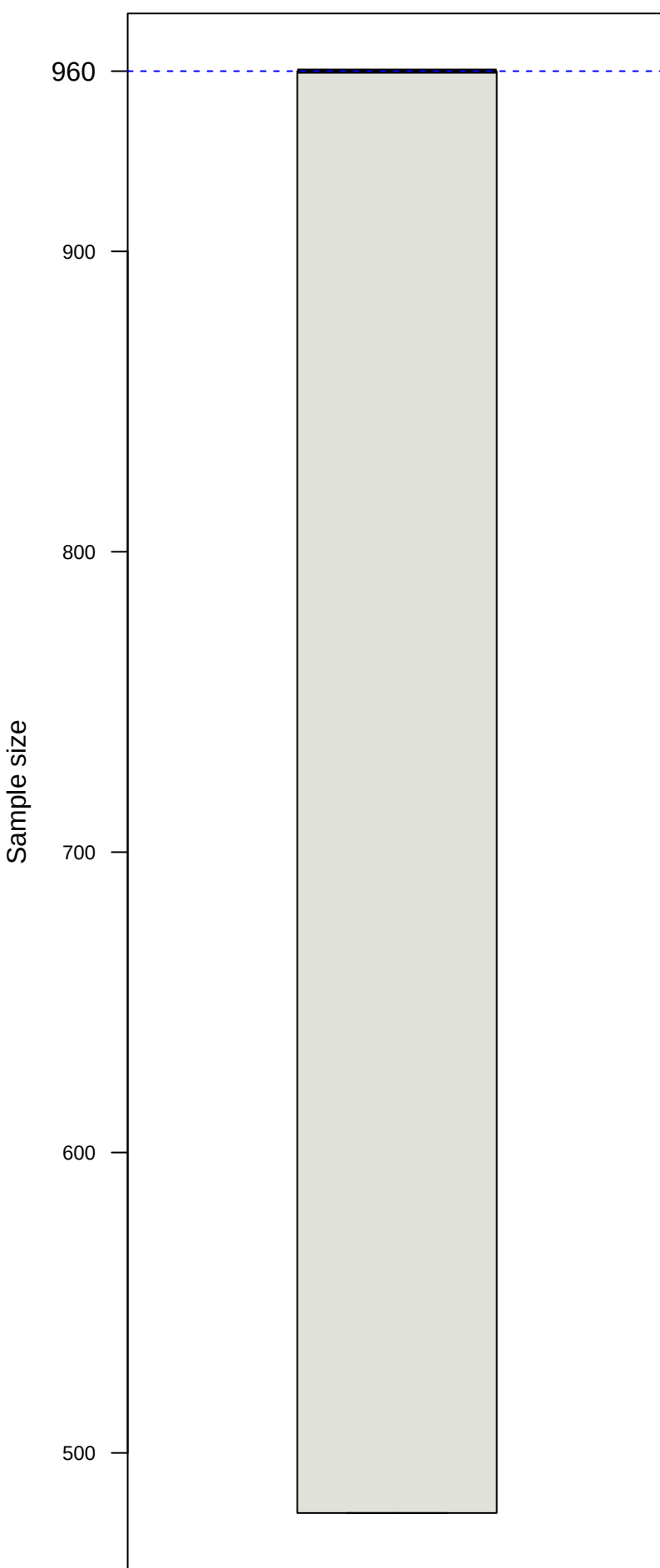
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

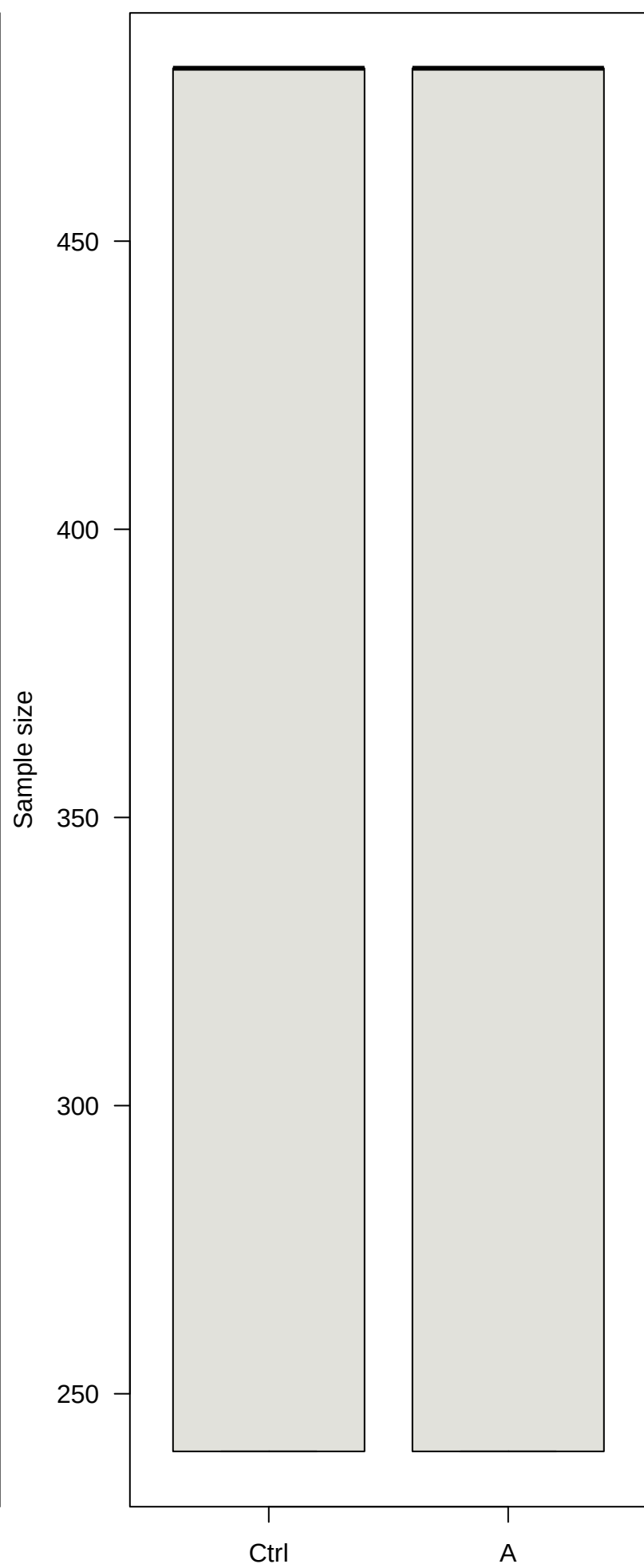
is overstated – see ../binary-binding for realistic ESS.

Config 104 – sample size (H1) [binary]

Total



Per group



Config 118 (0.05|0.9|1|2| 0.6|0.5| 0.7| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.700$

effect $\pi_t = 0.900$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9326

FWER (P reject ≥ 1 | H0) : 0.0501

Max sample size (planned) : 152

ESS under H1 (overstated) : 119.9

ESS under H0 (overstated) : 151.2

NB: design powered for the binary test (δ/σ), so achieved

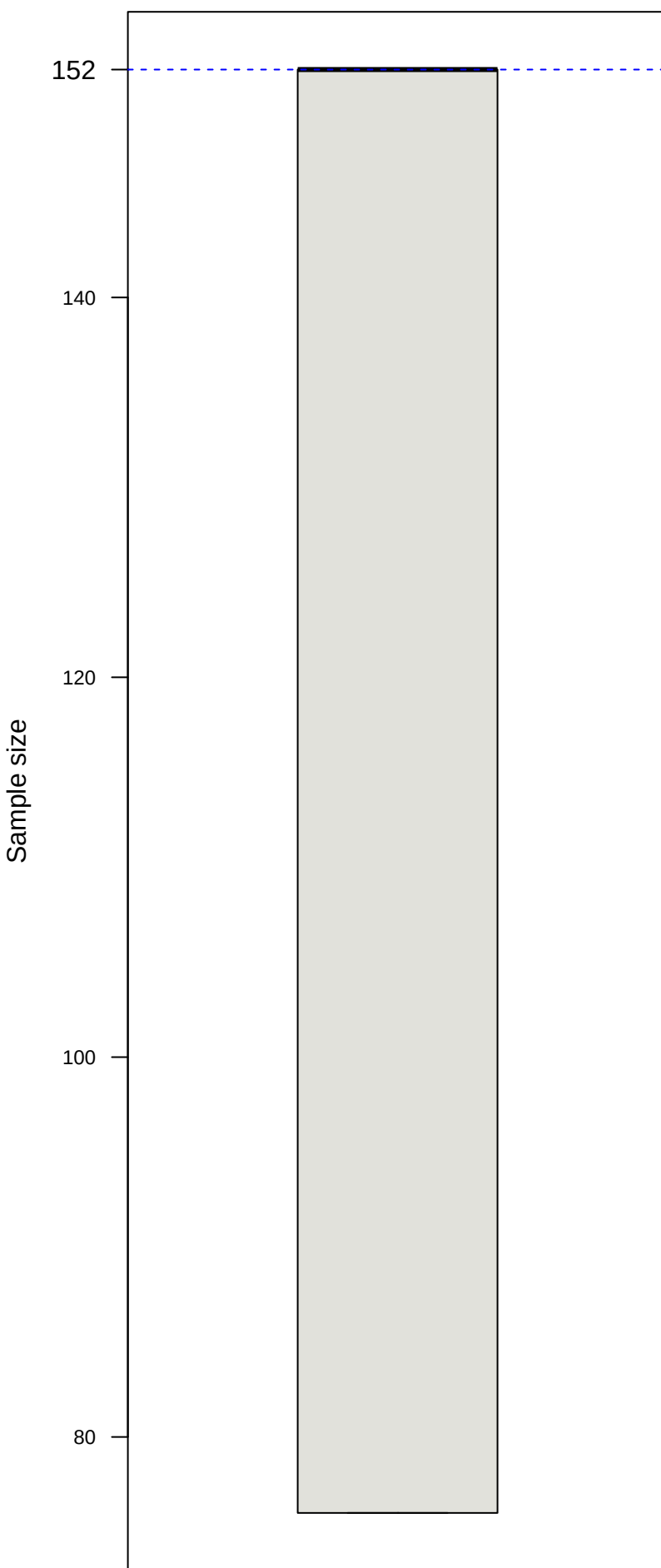
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

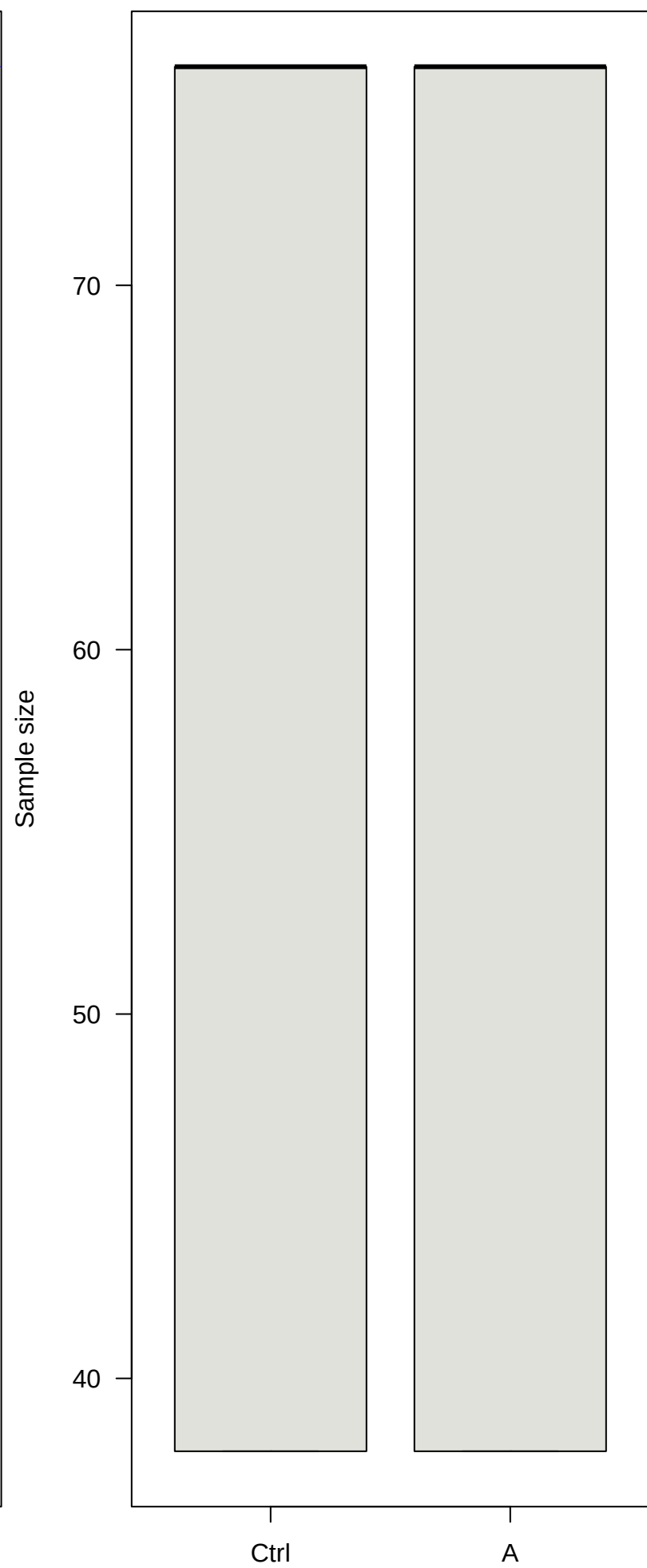
is overstated – see ../binary-binding for realistic ESS.

Config 118 – sample size (H1) [binary]

Total



Per group



Config 122 (0.05|0.9|1|2|0.65|0.5|0.05| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.050

effect pi_t = 0.350 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9441

FWER (P reject ≥ 1 | H0) : 0.0462

Max sample size (planned) : 64

ESS under H1 (overstated) : 50.1

ESS under H0 (overstated) : 64.0

NB: design powered for the binary test (delta/sd), so achieved

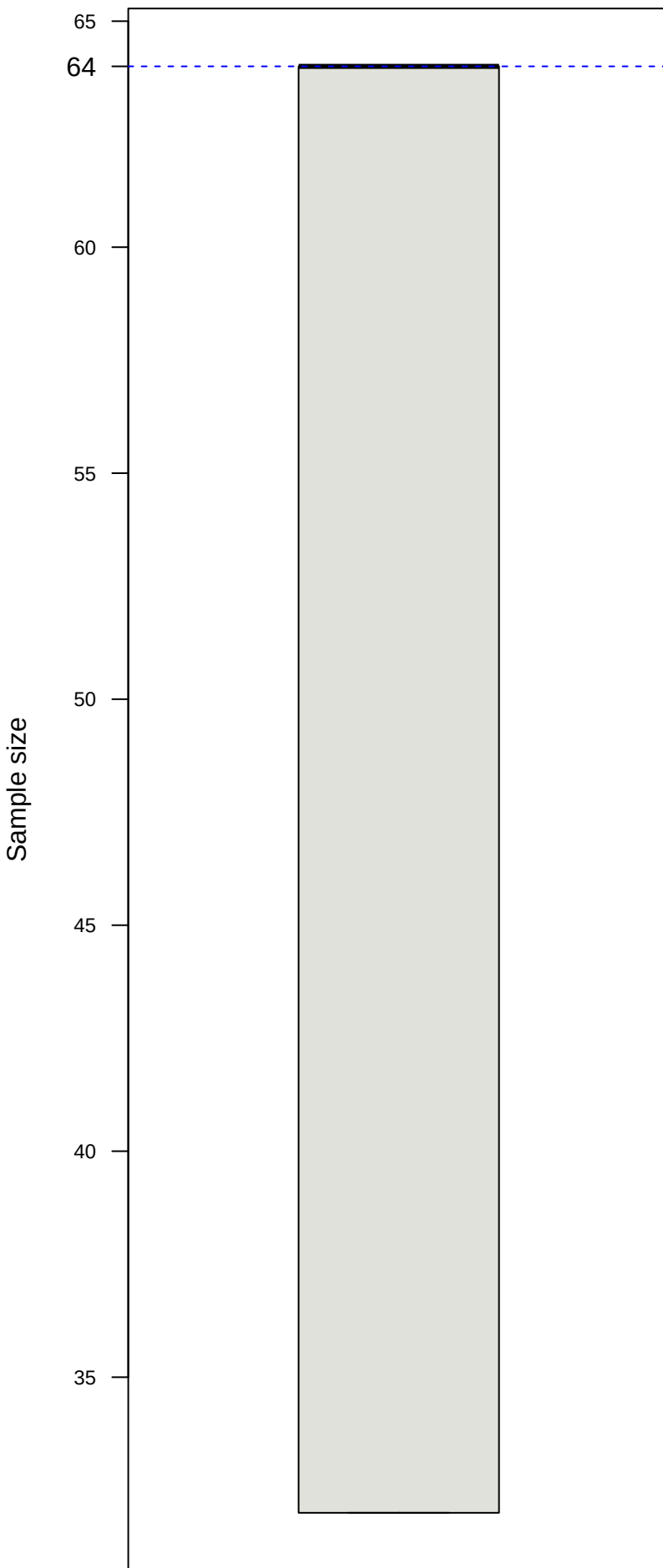
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

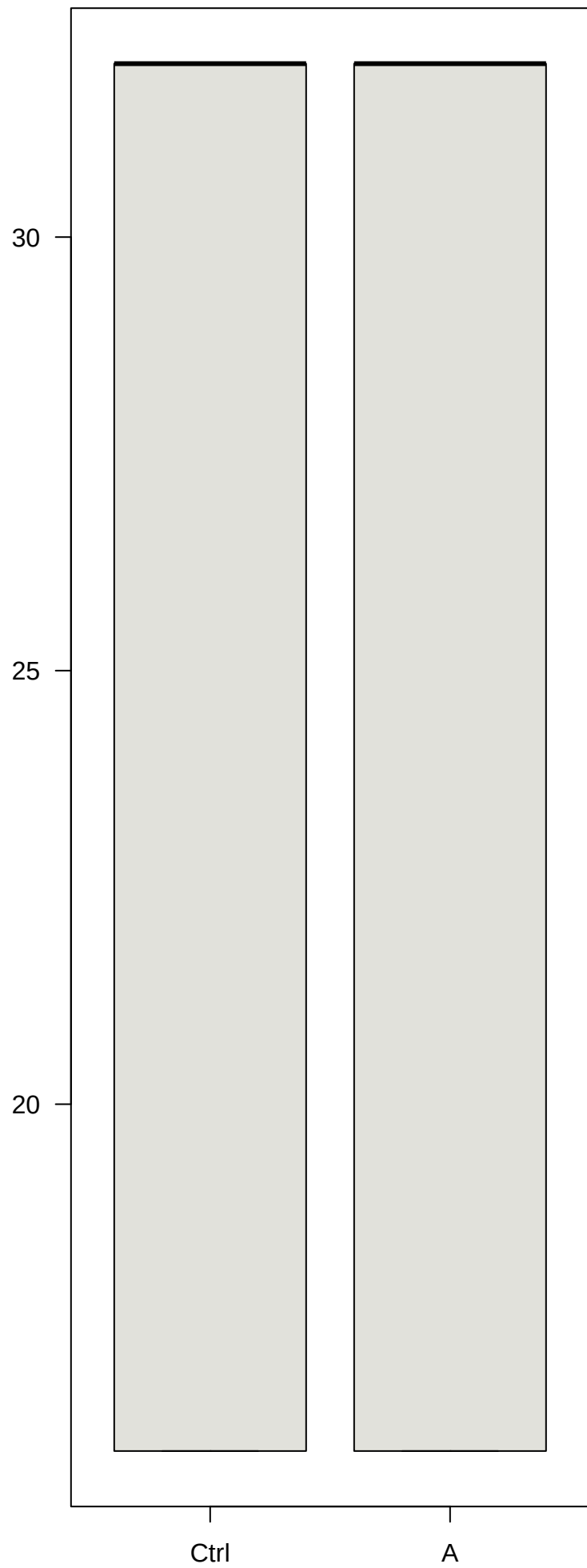
is overstated – see ../binary-binding for realistic ESS.

Config 122 – sample size (H1) [binary]

Total



Per group



Config 126 (0.05|0.9|1|2|0.65|0.5| 0.3| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9278

FWER (P reject ≥ 1 | H0) : 0.0516

Max sample size (planned) : 104

ESS under H1 (overstated) : 82.2

ESS under H0 (overstated) : 103.5

NB: design powered for the binary test (delta/sd), so achieved

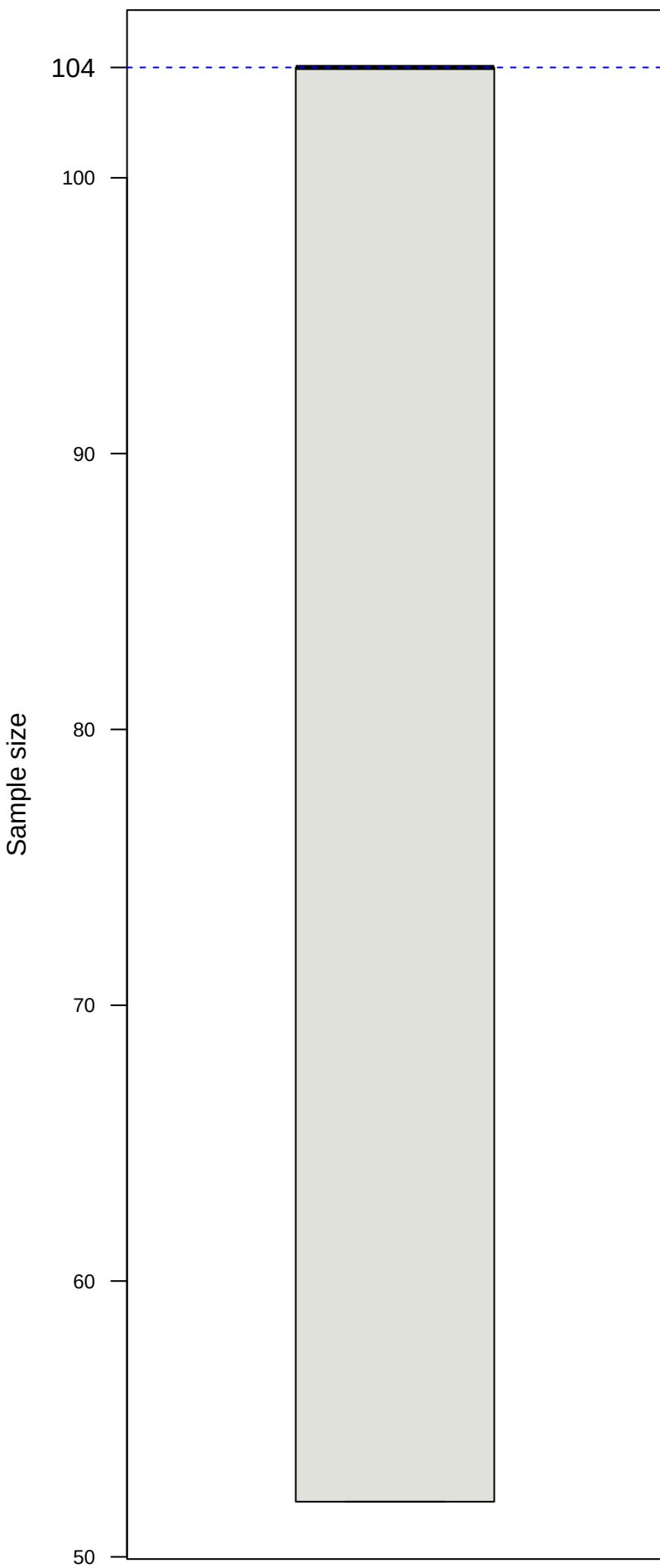
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

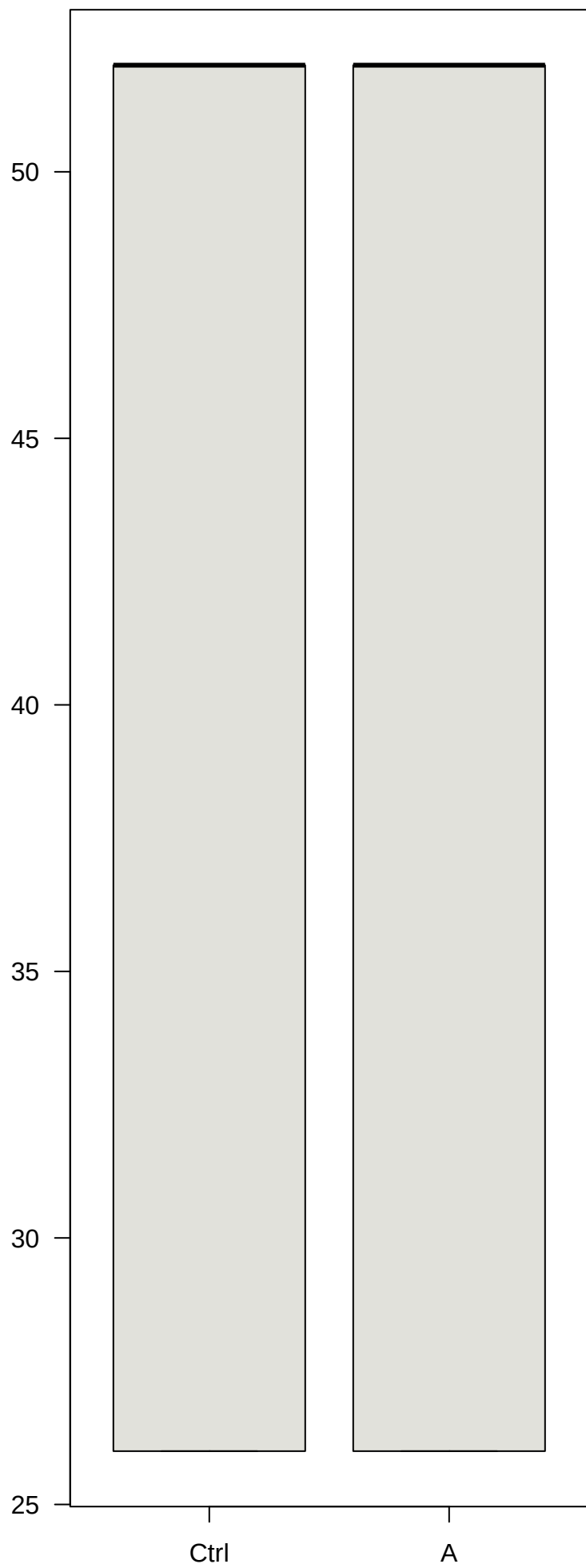
is overstated – see ../binary-binding for realistic ESS.

Config 126 – sample size (H1) [binary]

Total



Per group



Config 128 (0.05|0.9|1|2|0.65|0.5| 0.5| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9331

FWER (P reject ≥ 1 | H0) : 0.0437

Max sample size (planned) : 96

ESS under H1 (overstated) : 74.6

ESS under H0 (overstated) : 95.6

NB: design powered for the binary test (delta/sd), so achieved

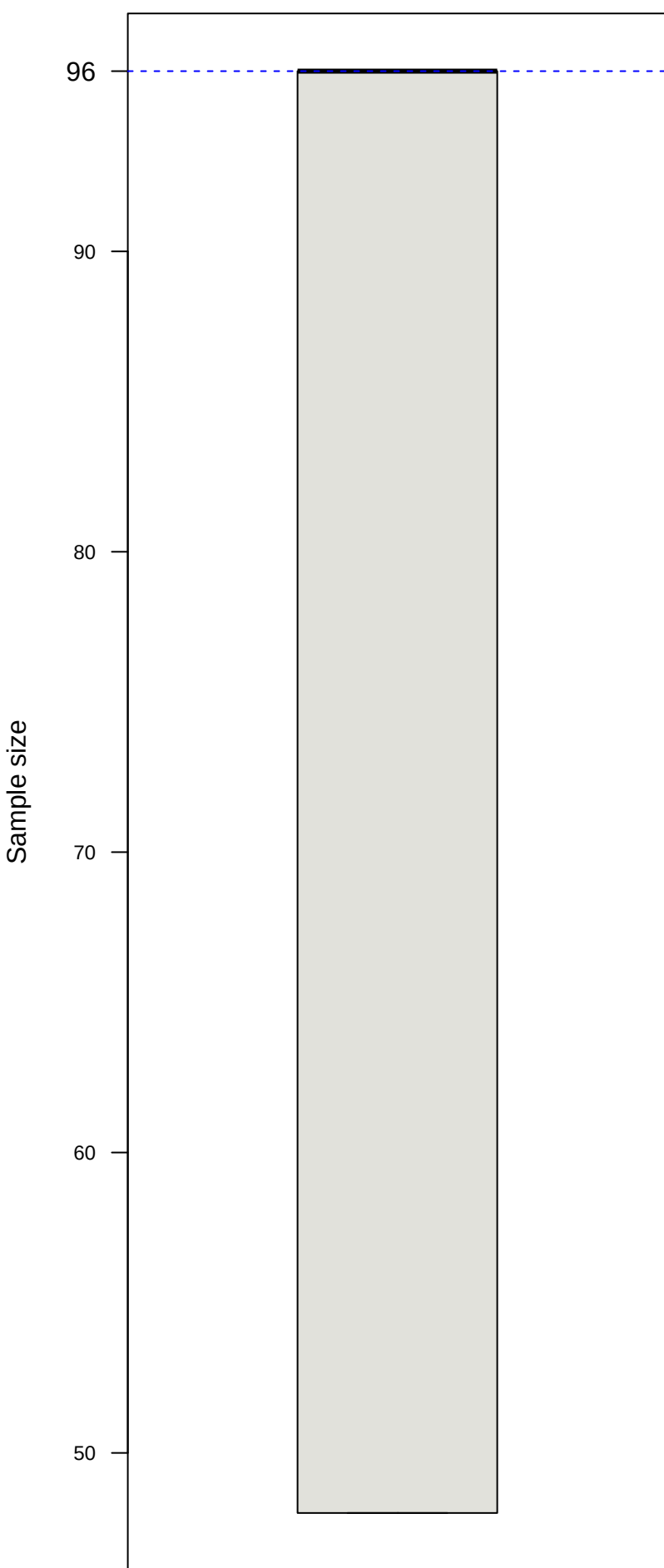
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

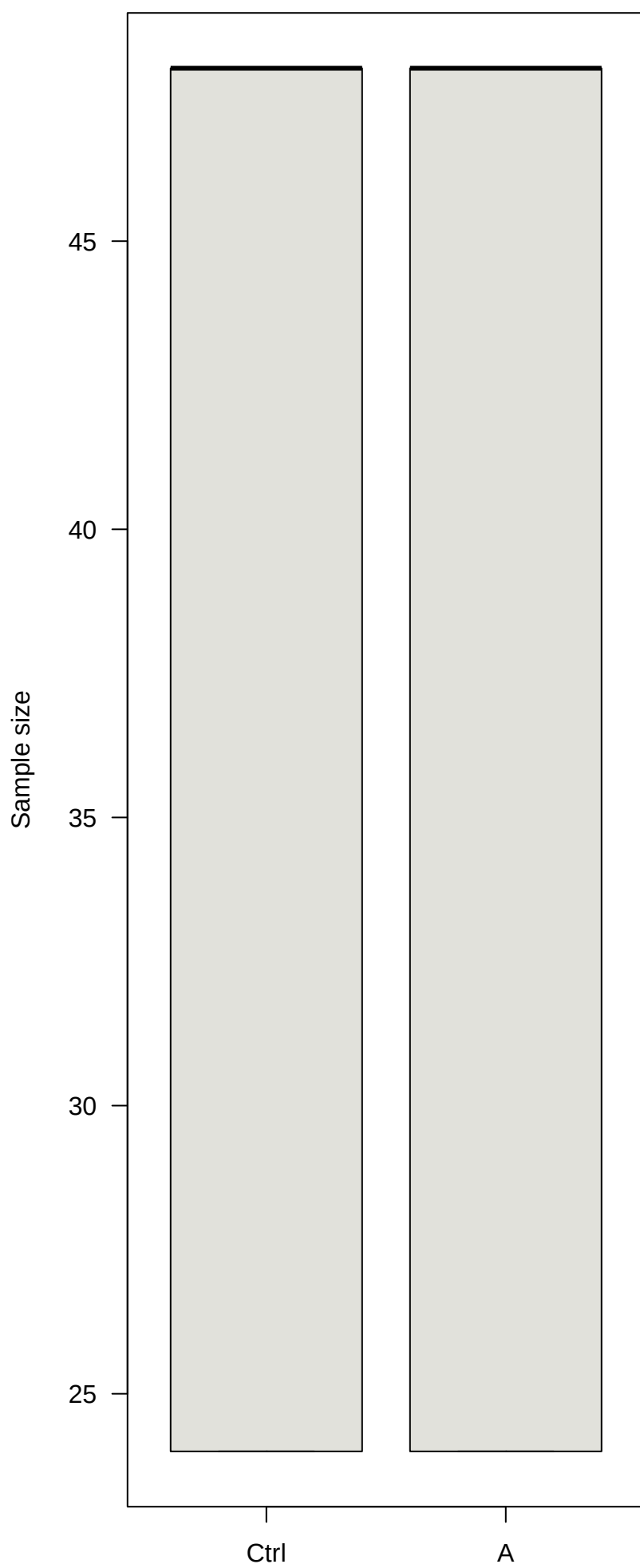
is overstated – see ../binary-binding for realistic ESS.

Config 128 – sample size (H1) [binary]

Total



Per group



Config 138 (0.05|0.9|1|2|0.75|0.5| 0.3| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9400

FWER (P reject ≥ 1 | H0) : 0.0540

Max sample size (planned) : 36

ESS under H1 (overstated) : 28.2

ESS under H0 (overstated) : 35.9

NB: design powered for the binary test (delta/sd), so achieved

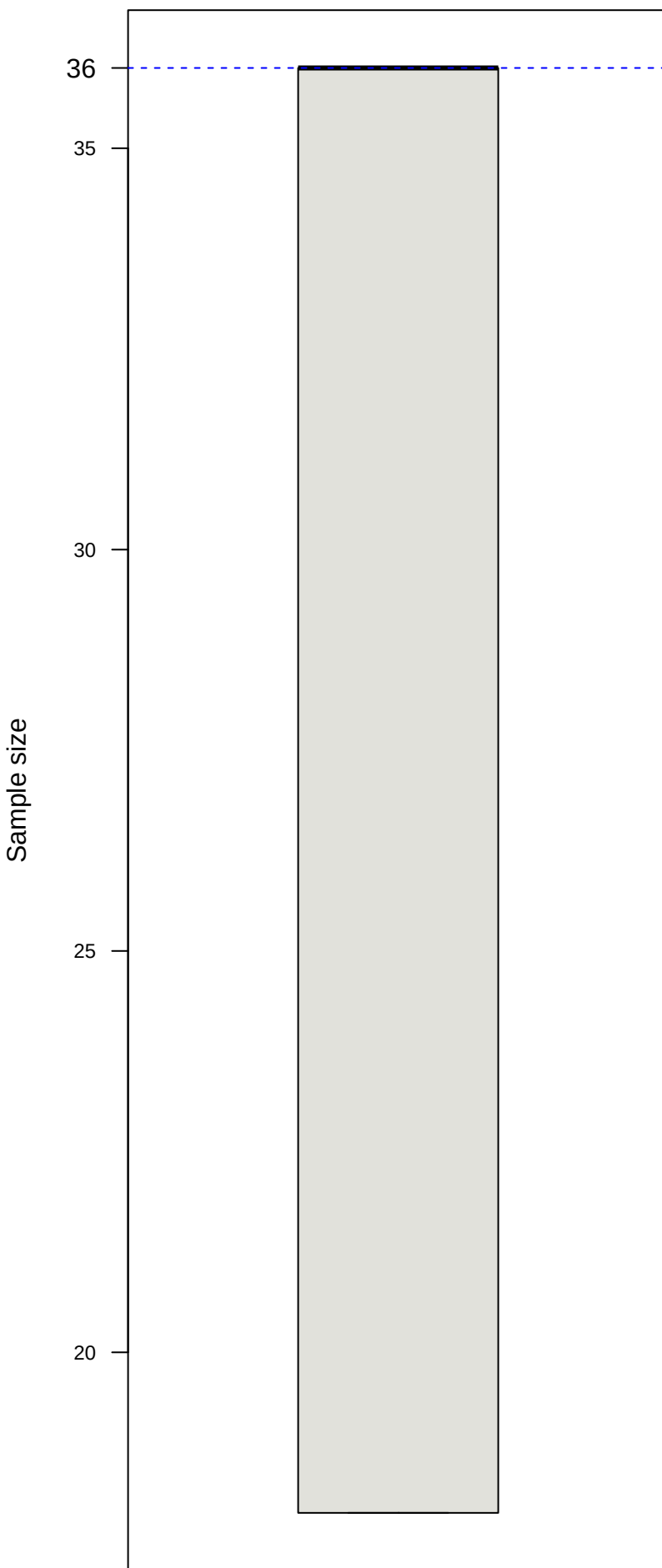
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

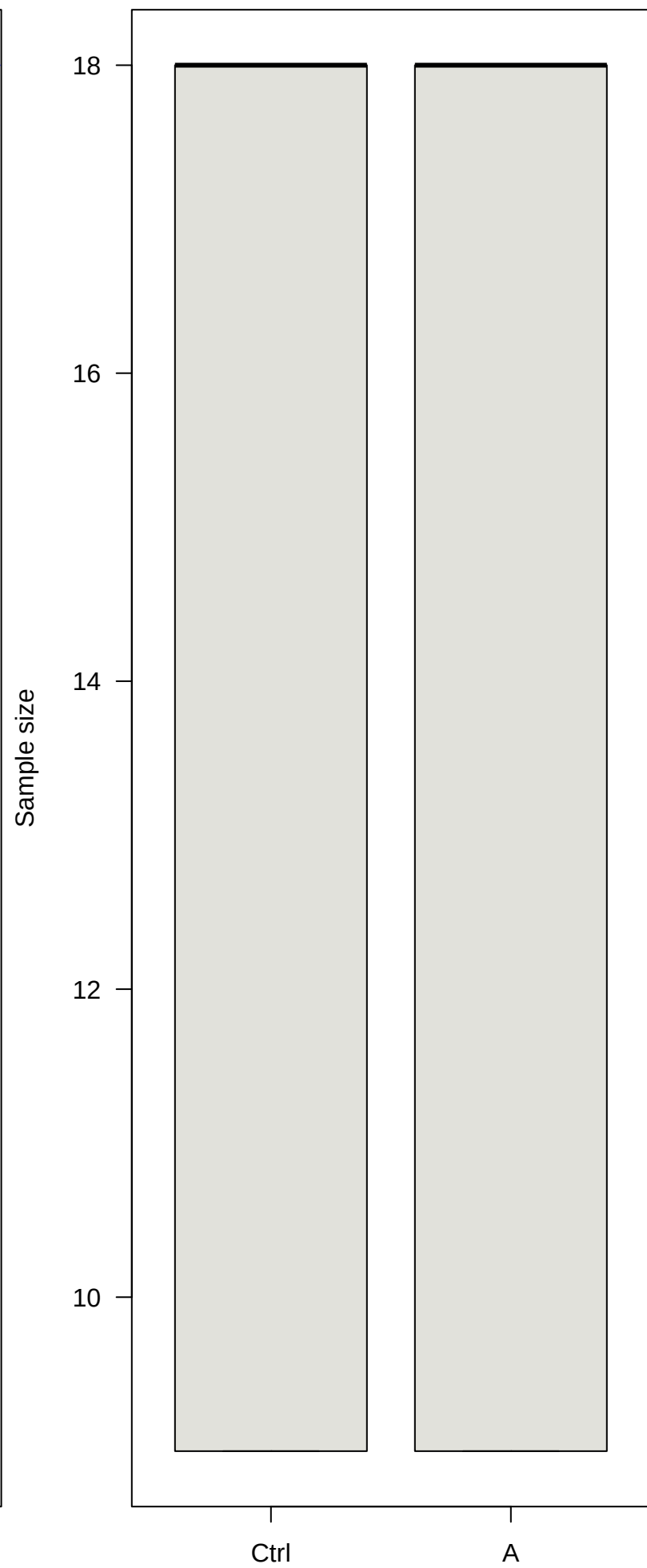
is overstated – see ../binary-binding for realistic ESS.

Config 138 – sample size (H1) [binary]

Total



Per group



Config 148 (0.05|0.9|1|2|0.85|0.5| 0.1| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.100

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9237

FWER (P reject ≥ 1 | H0) : 0.0170

Max sample size (planned) : 16

ESS under H1 (overstated) : 13.9

ESS under H0 (overstated) : 16.0

NB: design powered for the binary test (delta/sd), so achieved

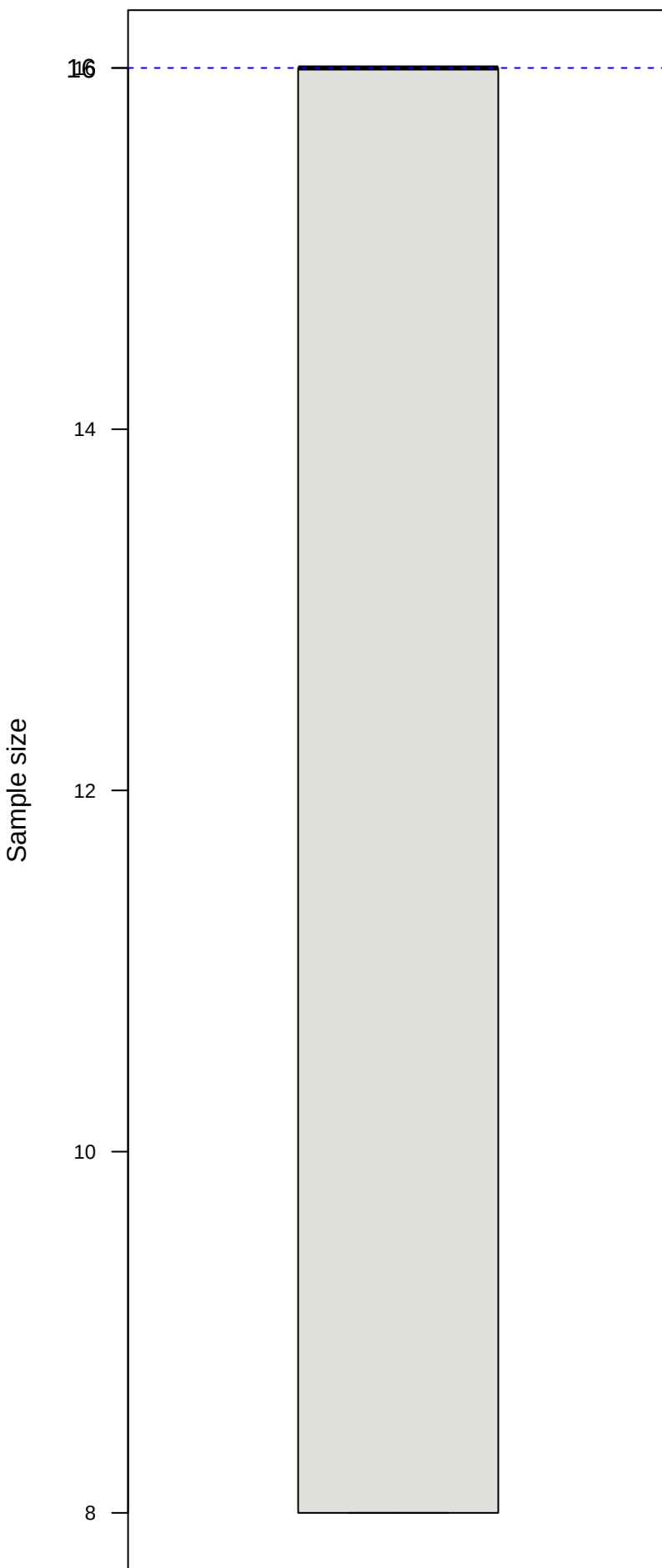
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

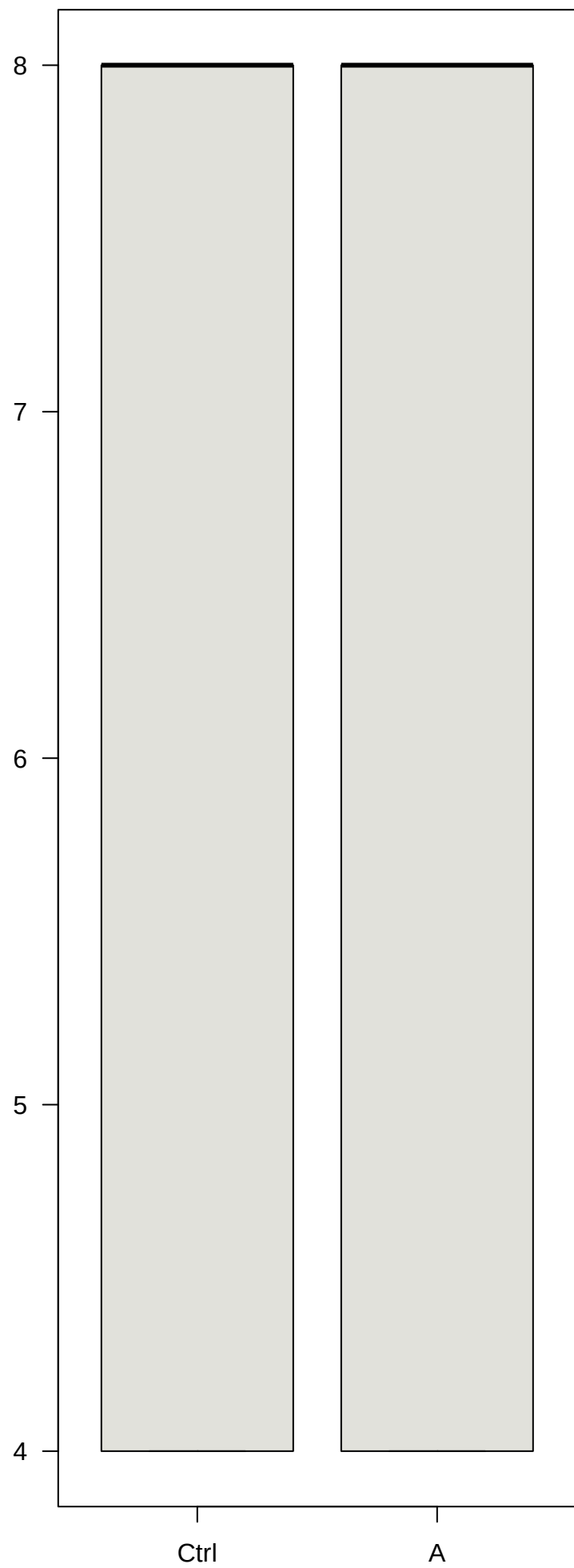
is overstated – see ../binary-binding for realistic ESS.

Config 148 – sample size (H1) [binary]

Total



Per group



Config 158 (0.05|0.9|1|2|0.95|0.5|0.05| sep) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.050

effect pi_t = 0.950 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9424

FWER (P reject ≥ 1 | H0) : 0.0004

Max sample size (planned) : 8

ESS under H1 (overstated) : 8.0

ESS under H0 (overstated) : 8.0

NB: design powered for the binary test (delta/sd), so achieved

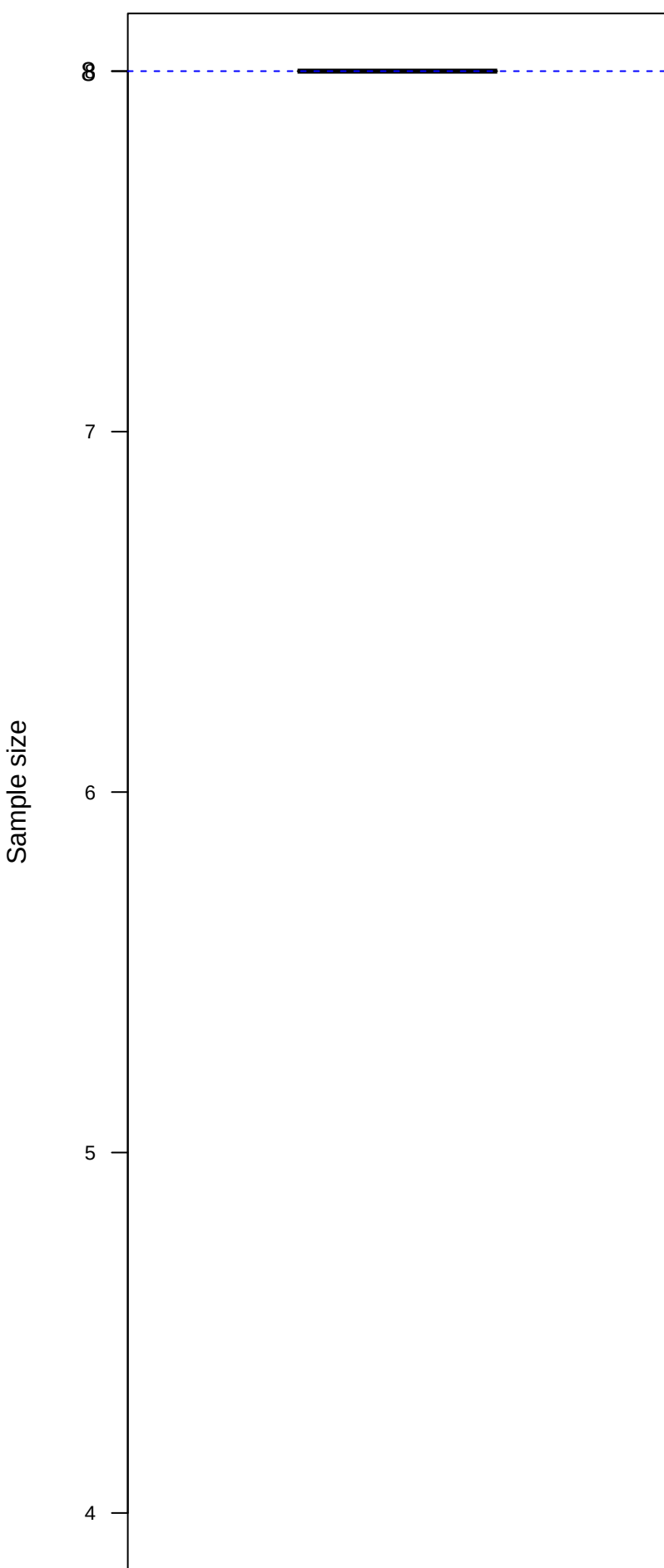
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

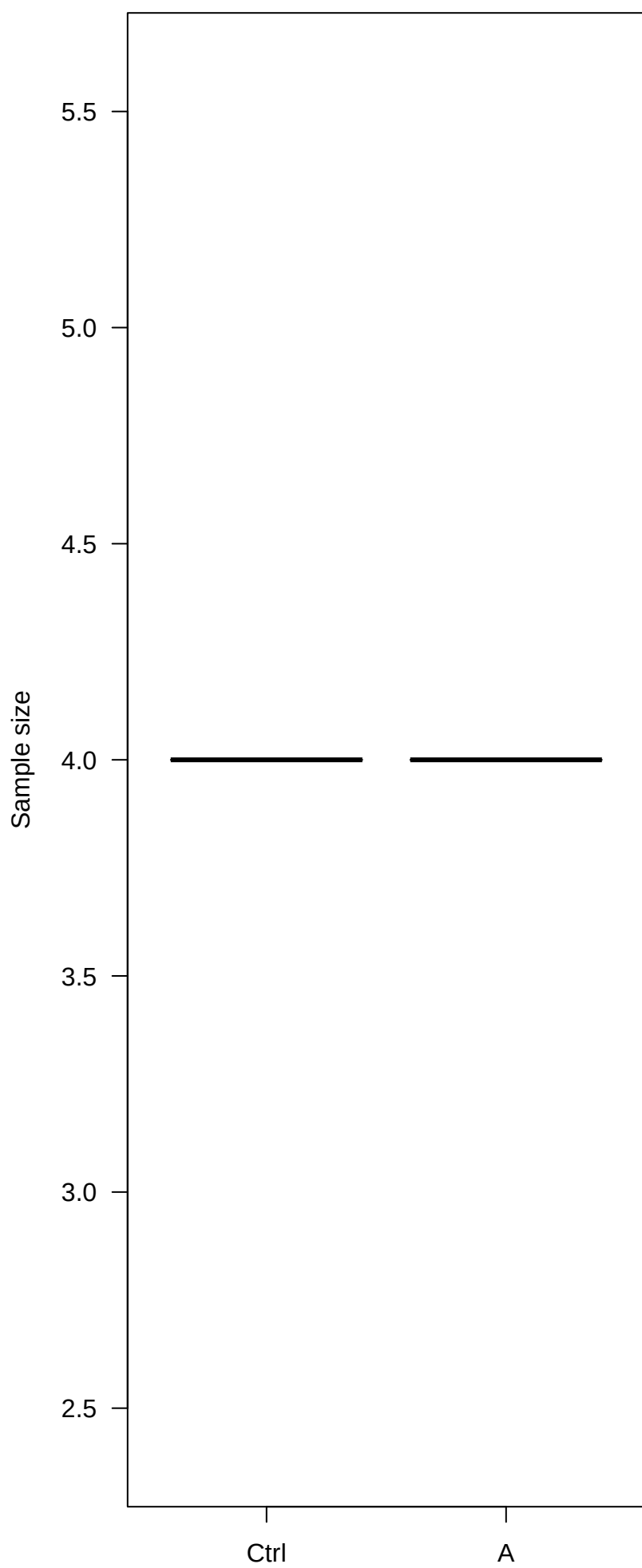
is overstated – see ../binary-binding for realistic ESS.

Config 158 – sample size (H1) [binary]

Total



Per group



Config 178 (0.05|0.9|4|1|0.52|0.5| 0.7| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.740 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8954

FWER (P reject ≥ 1 | H0) : 0.0508

Max sample size (planned) : 14920

ESS under H1 (overstated) : 14920.0

ESS under H0 (overstated) : 14920.0

NB: design powered for the binary test (delta/sd), so achieved

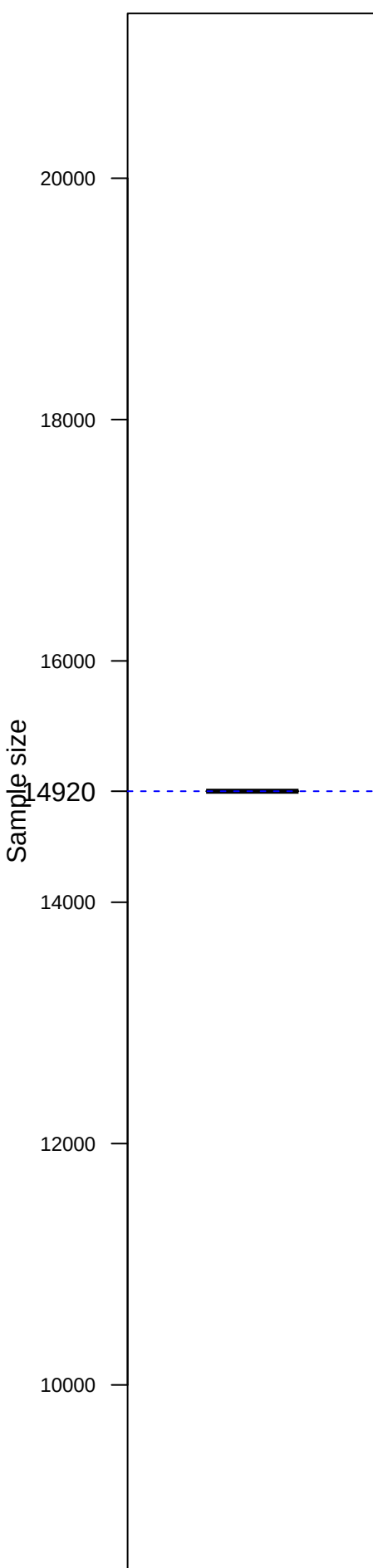
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

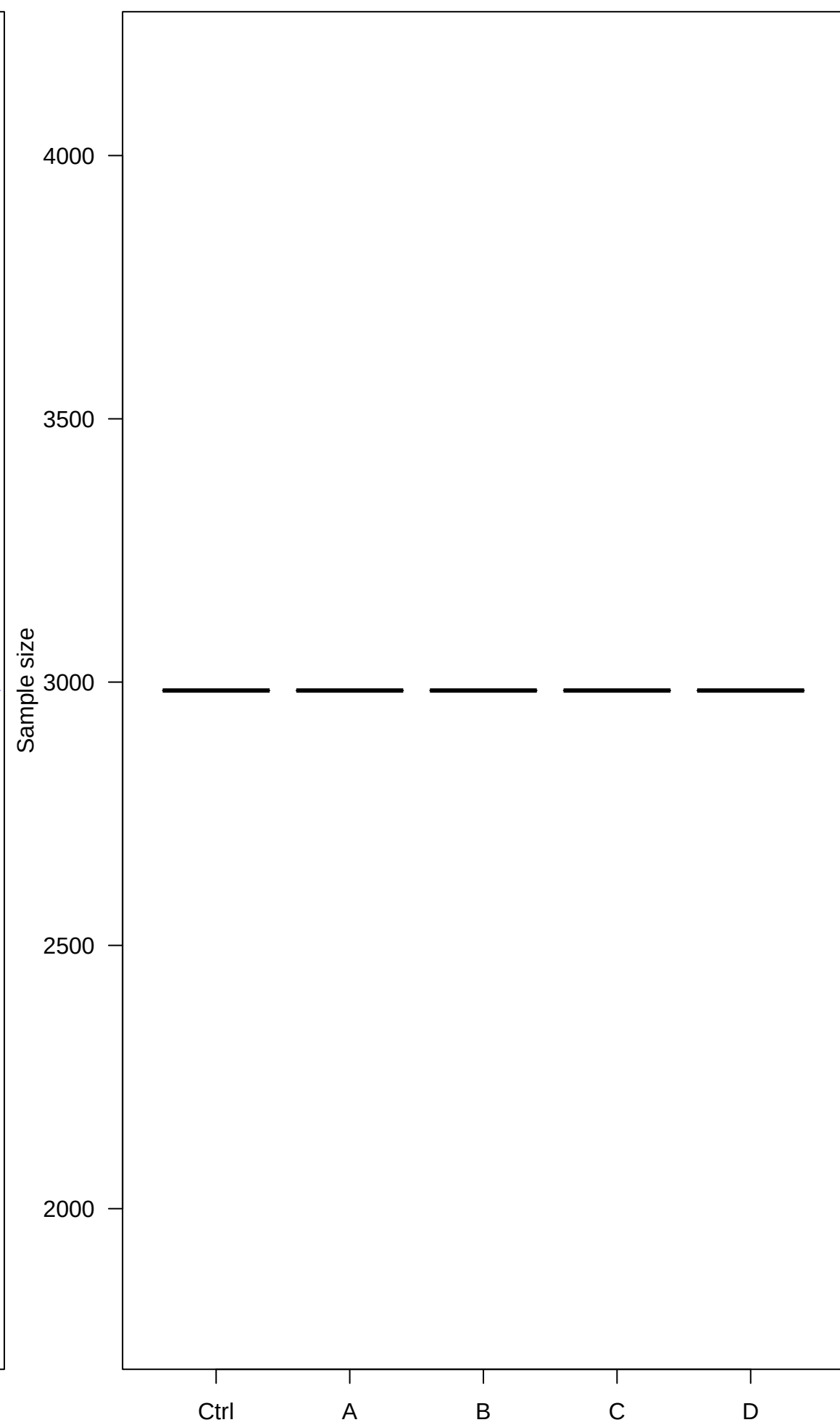
is overstated – see ../binary-binding for realistic ESS.

Config 178 – sample size (H1) [binary]

Total



Per group



Config 180 (0.05|0.9|4|1|0.52|0.5| 0.9| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.900

effect pi_t = 0.940 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8767

FWER (P reject ≥ 1 | H0) : 0.0496

Max sample size (planned) : 5440

ESS under H1 (overstated) : 5440.0

ESS under H0 (overstated) : 5440.0

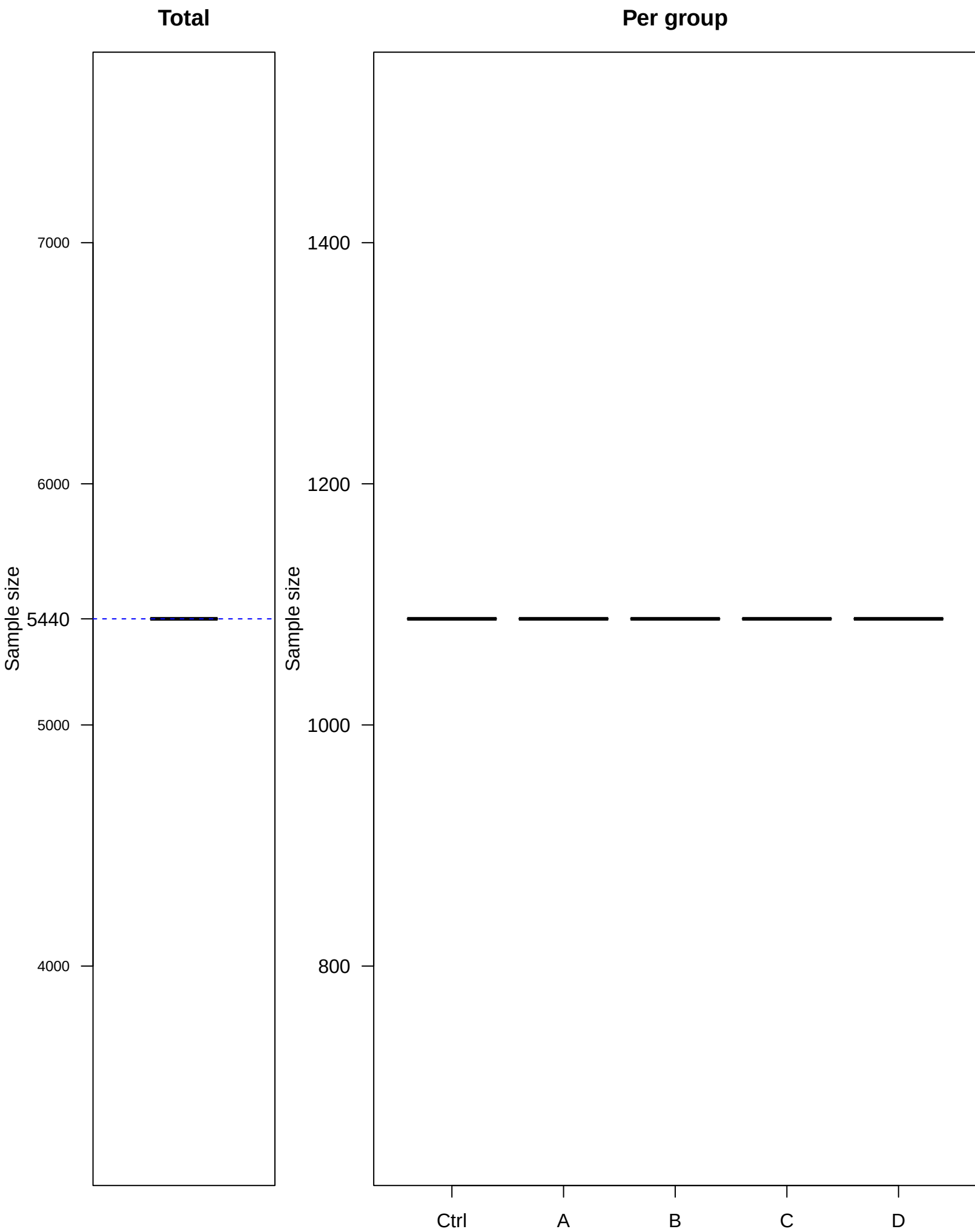
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 180 – sample size (H1) [binary]



Config 188 (0.05|0.9|4|1|0.55|0.5| 0.5| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9016

FWER (P reject ≥ 1 | H0) : 0.0540

Max sample size (planned) : 2925

ESS under H1 (overstated) : 2925.0

ESS under H0 (overstated) : 2925.0

NB: design powered for the binary test (delta/sd), so achieved

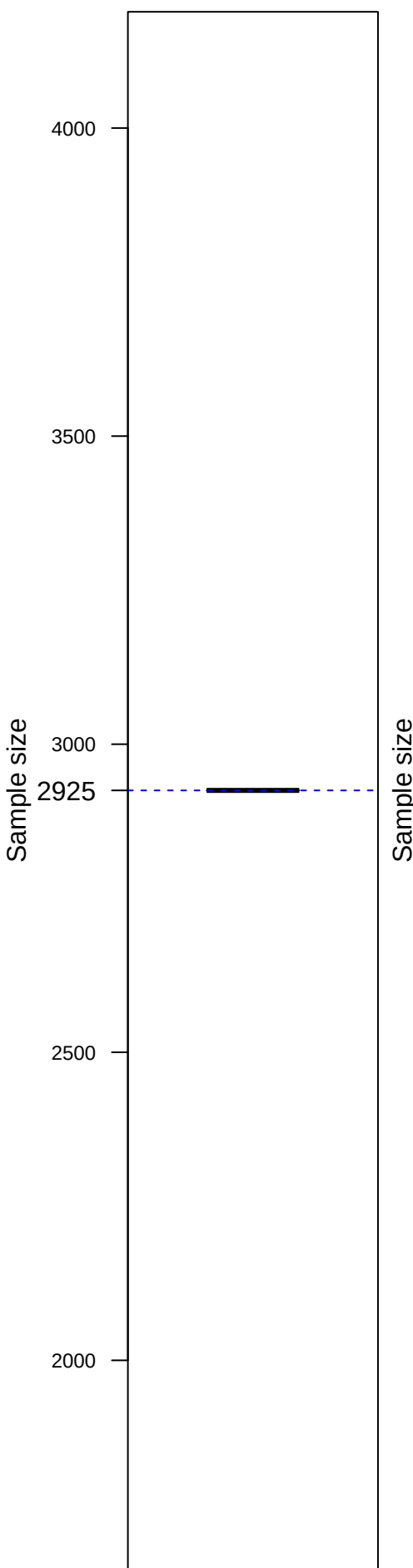
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

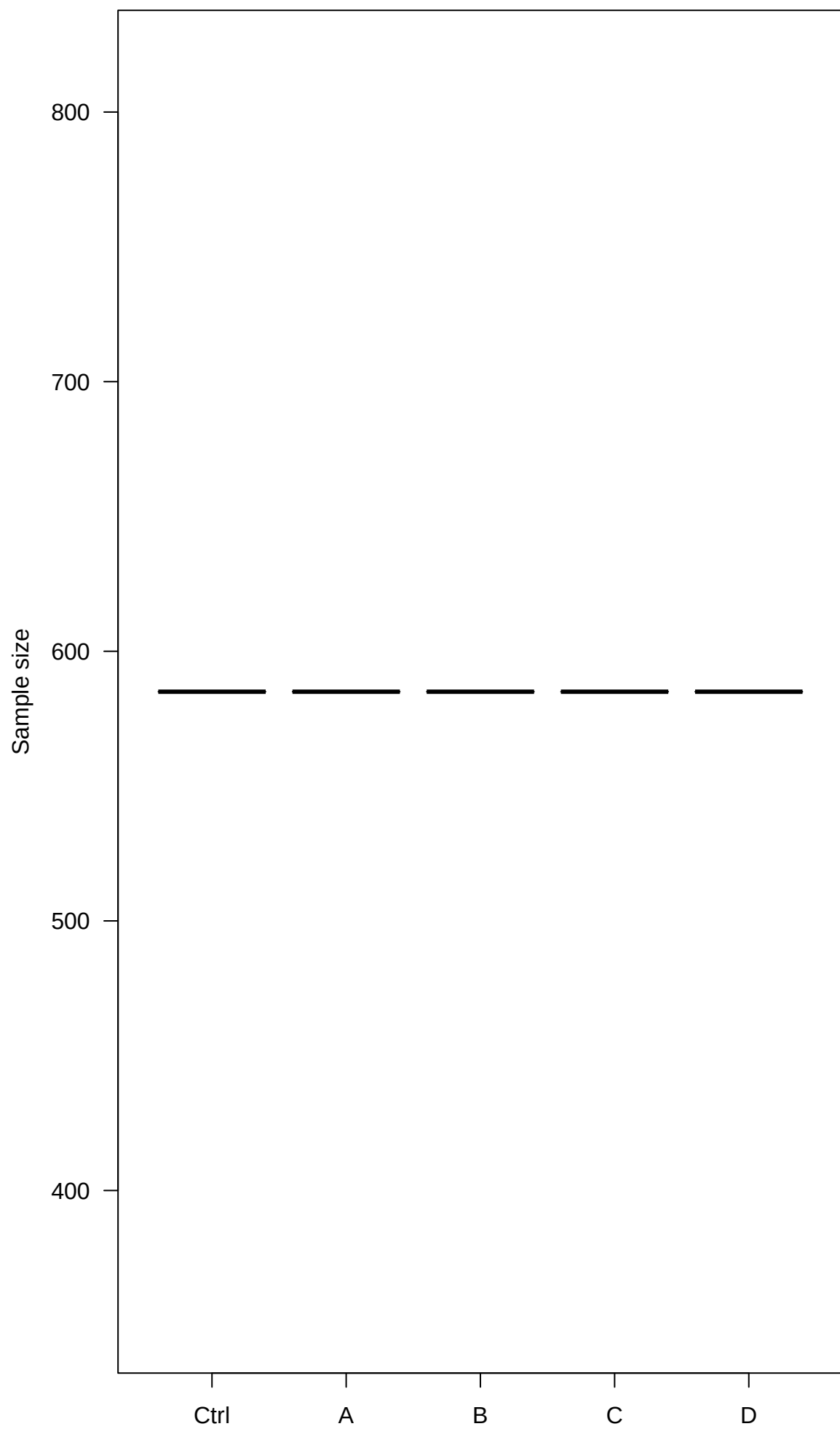
is overstated – see ../binary-binding for realistic ESS.

Config 188 – sample size (H1) [binary]

Total



Per group



Config 202 (0.05|0.9|4|1| 0.6|0.5| 0.7| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.900 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8600

FWER (P reject ≥ 1 | H0) : 0.0478

Max sample size (planned) : 465

ESS under H1 (overstated) : 465.0

ESS under H0 (overstated) : 465.0

NB: design powered for the binary test (delta/sd), so achieved

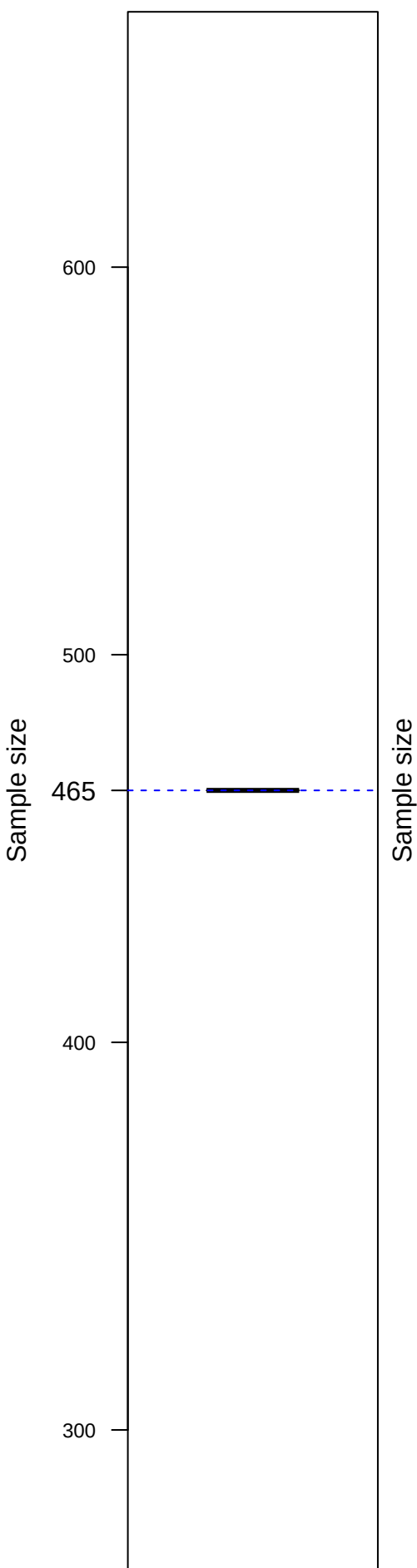
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

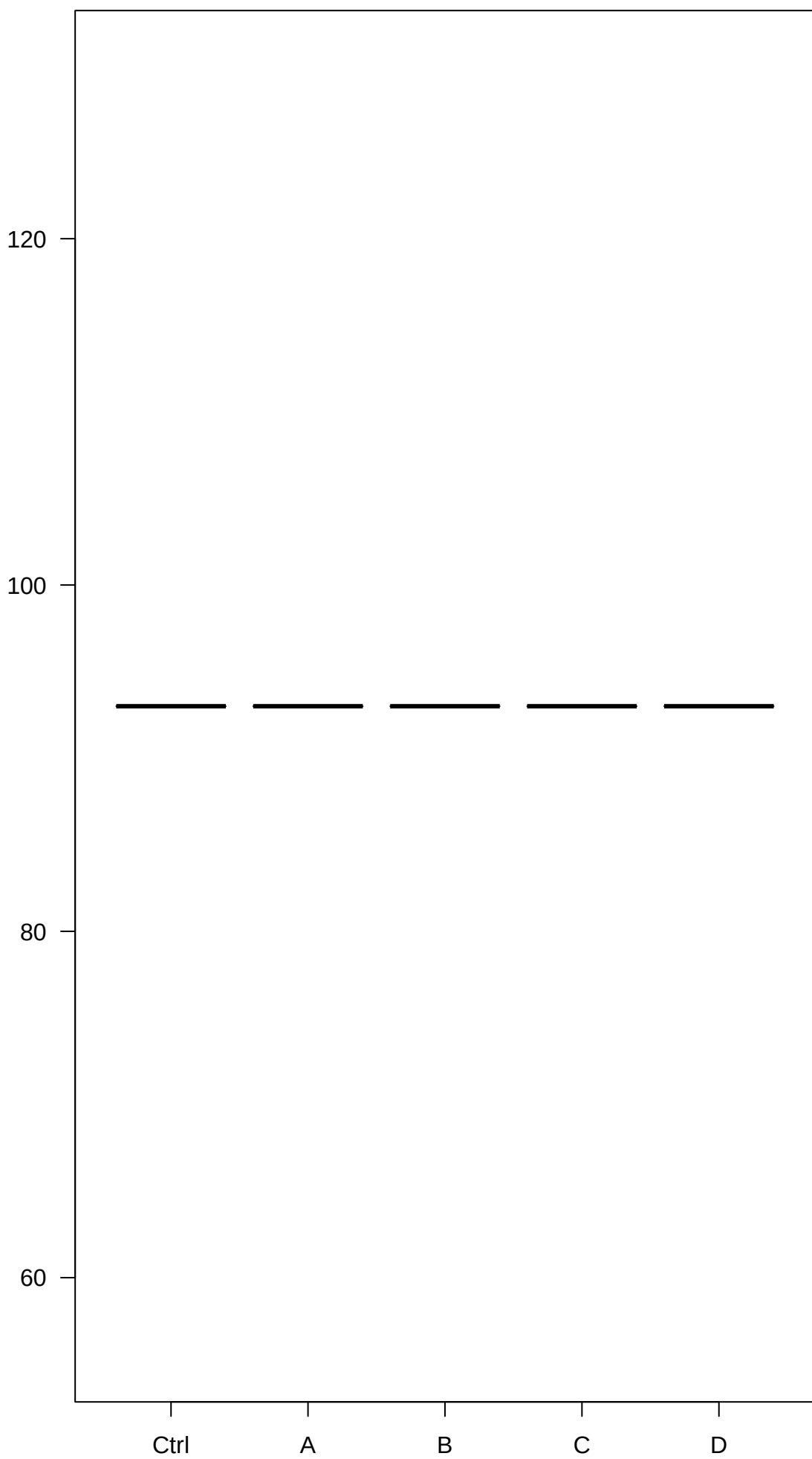
is overstated – see ../binary-binding for realistic ESS.

Config 202 – sample size (H1) [binary]

Total



Per group



Config 206 (0.05|0.9|4|1|0.65|0.5|0.05| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.050

effect pi_t = 0.350 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9747

FWER (P reject ≥ 1 | H0) : 0.0528

Max sample size (planned) : 200

ESS under H1 (overstated) : 200.0

ESS under H0 (overstated) : 200.0

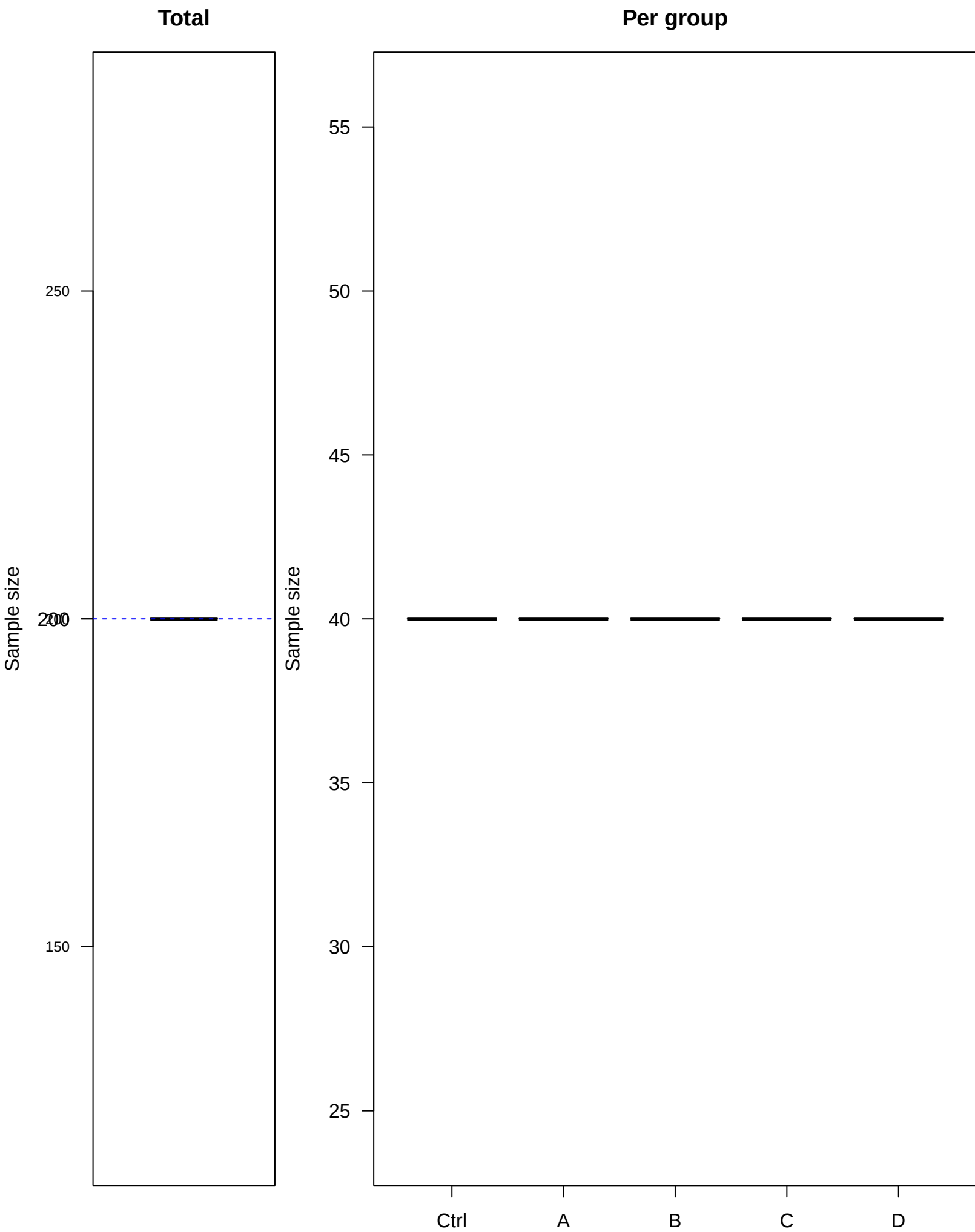
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 206 – sample size (H1) [binary]



Config 210 (0.05|0.9|4|1|0.65|0.5| 0.3| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9155

FWER (P reject ≥ 1 | H0) : 0.0496

Max sample size (planned) : 315

ESS under H1 (overstated) : 315.0

ESS under H0 (overstated) : 315.0

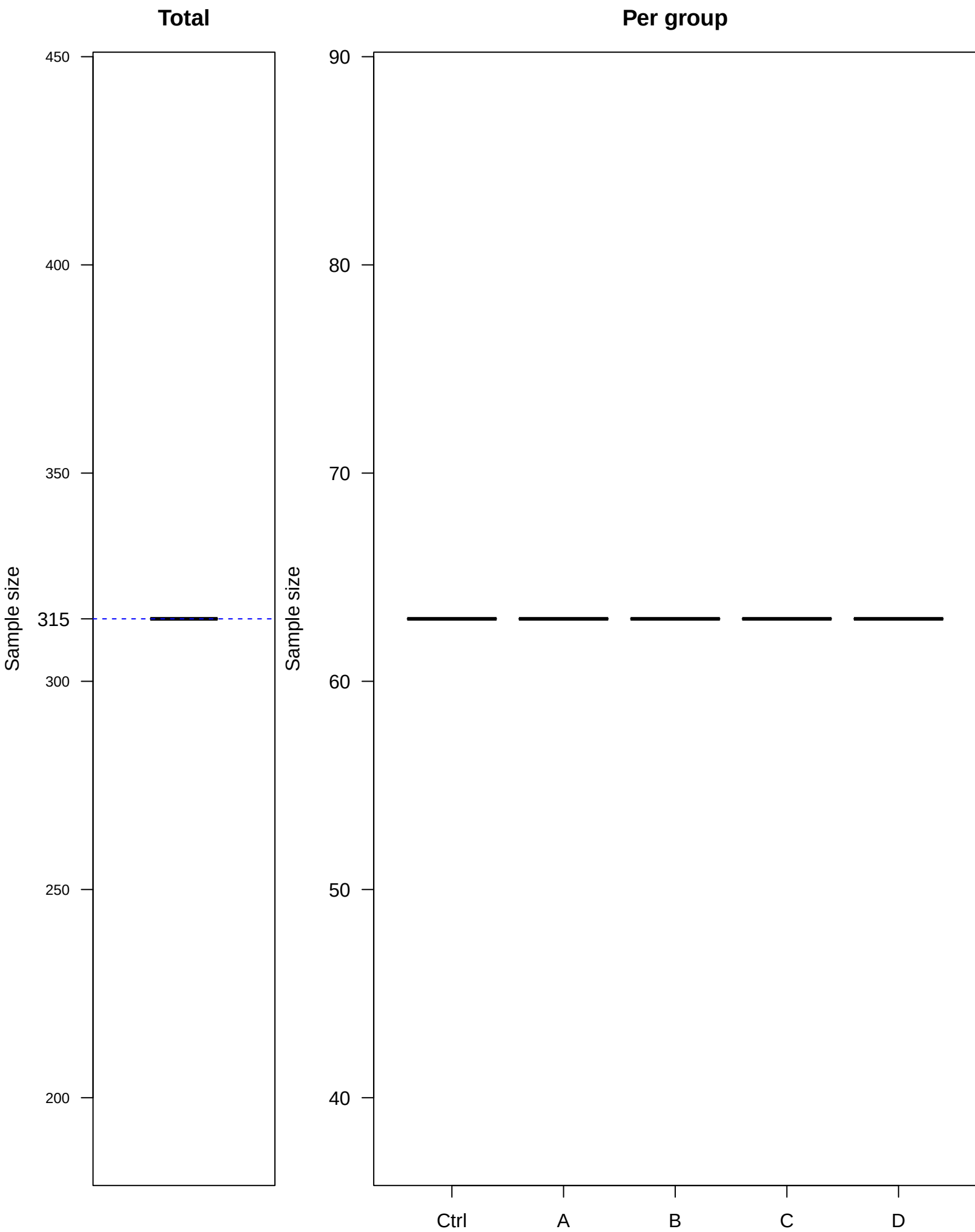
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 210 – sample size (H1) [binary]



Config 212 (0.05|0.9|4|1|0.65|0.5| 0.5| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8868

FWER (P reject ≥ 1 | H0) : 0.0509

Max sample size (planned) : 290

ESS under H1 (overstated) : 290.0

ESS under H0 (overstated) : 290.0

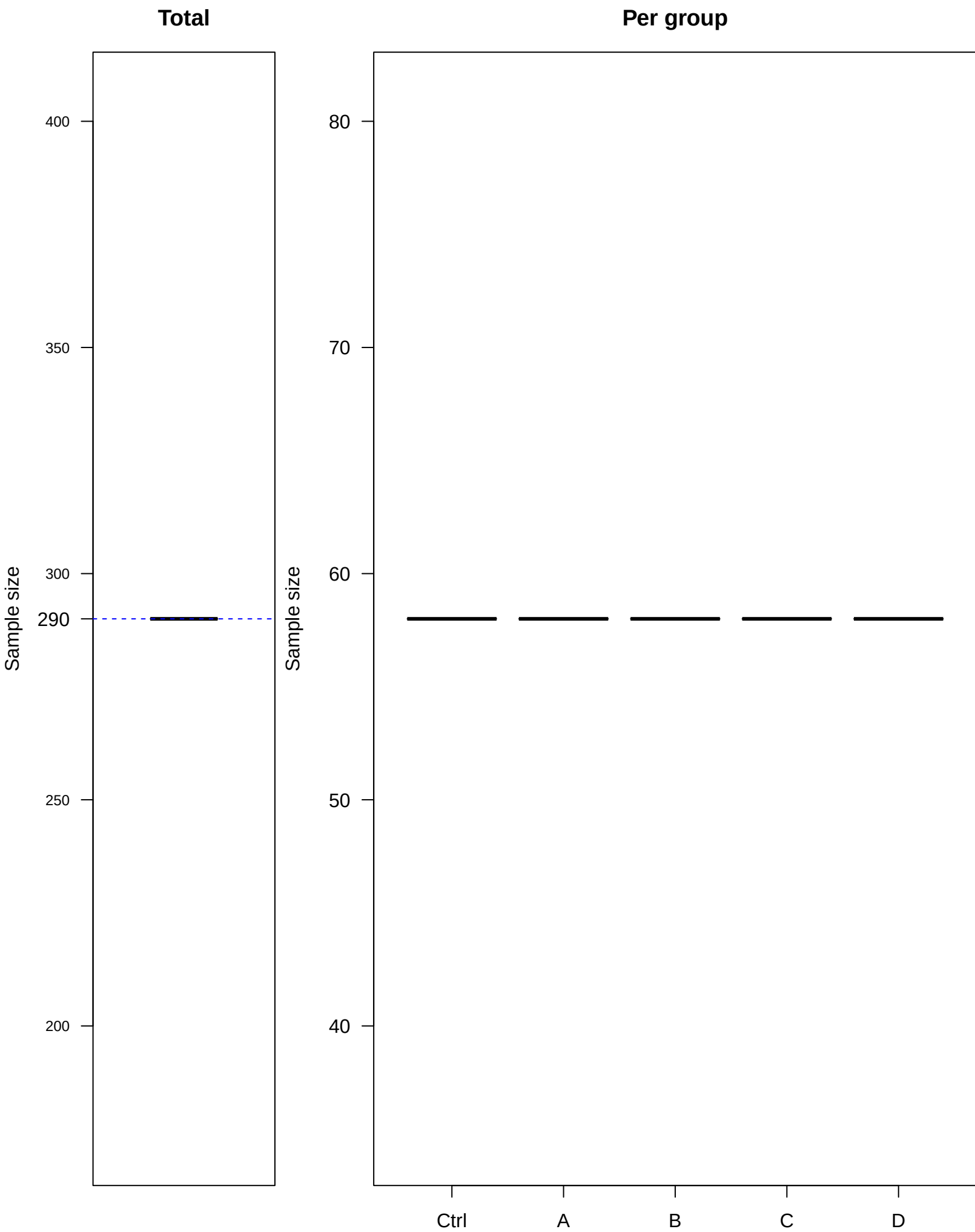
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 212 – sample size (H1) [binary]



Config 222 (0.05|0.9|4|1|0.75|0.5| 0.3| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9054

FWER (P reject ≥ 1 | H0) : 0.0562

Max sample size (planned) : 110

ESS under H1 (overstated) : 110.0

ESS under H0 (overstated) : 110.0

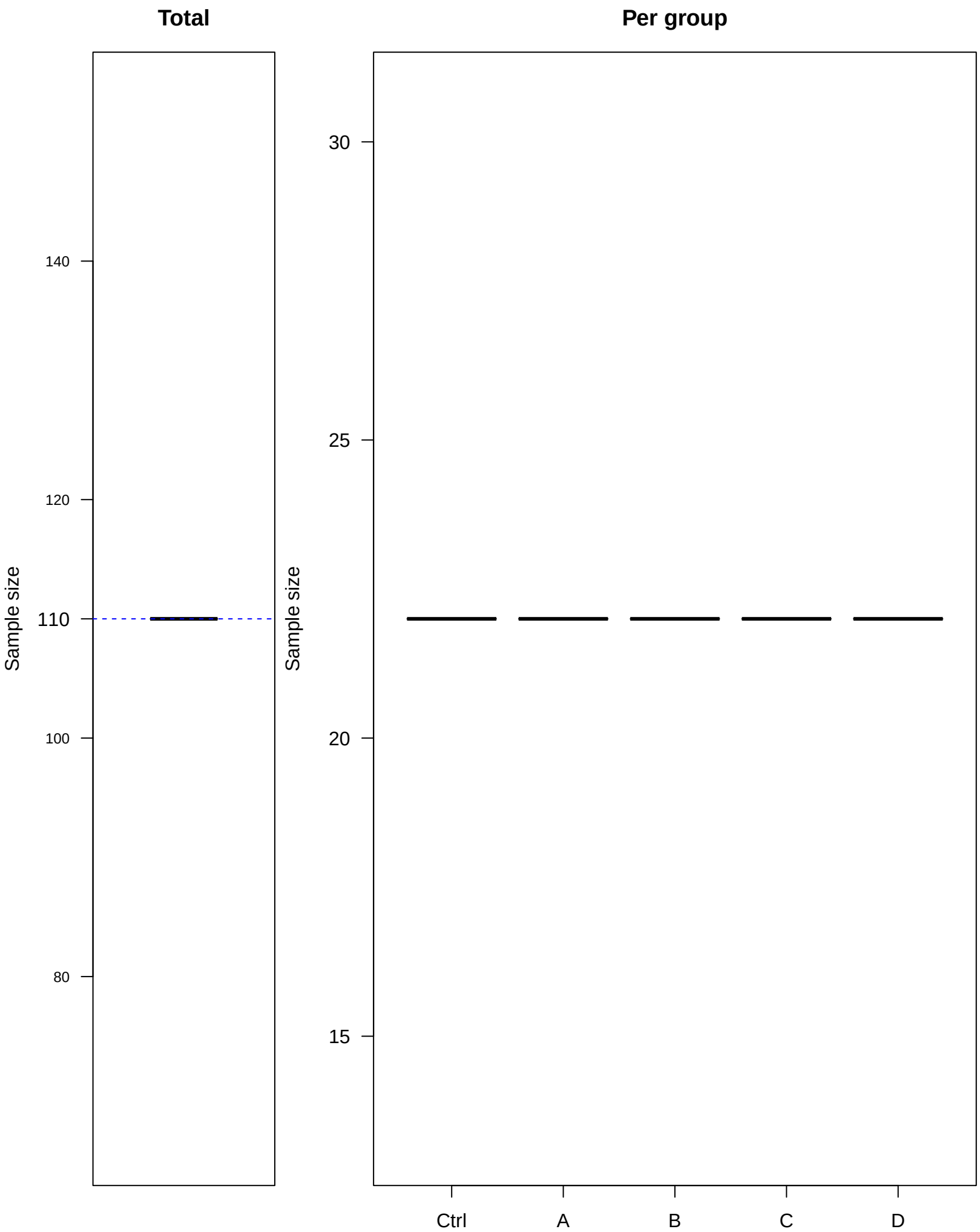
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 222 – sample size (H1) [binary]



Config 232 (0.05|0.9|4|1|0.85|0.5| 0.1| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.100

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9524

FWER (P reject ≥ 1 | H0) : 0.0652

Max sample size (planned) : 50

ESS under H1 (overstated) : 50.0

ESS under H0 (overstated) : 50.0

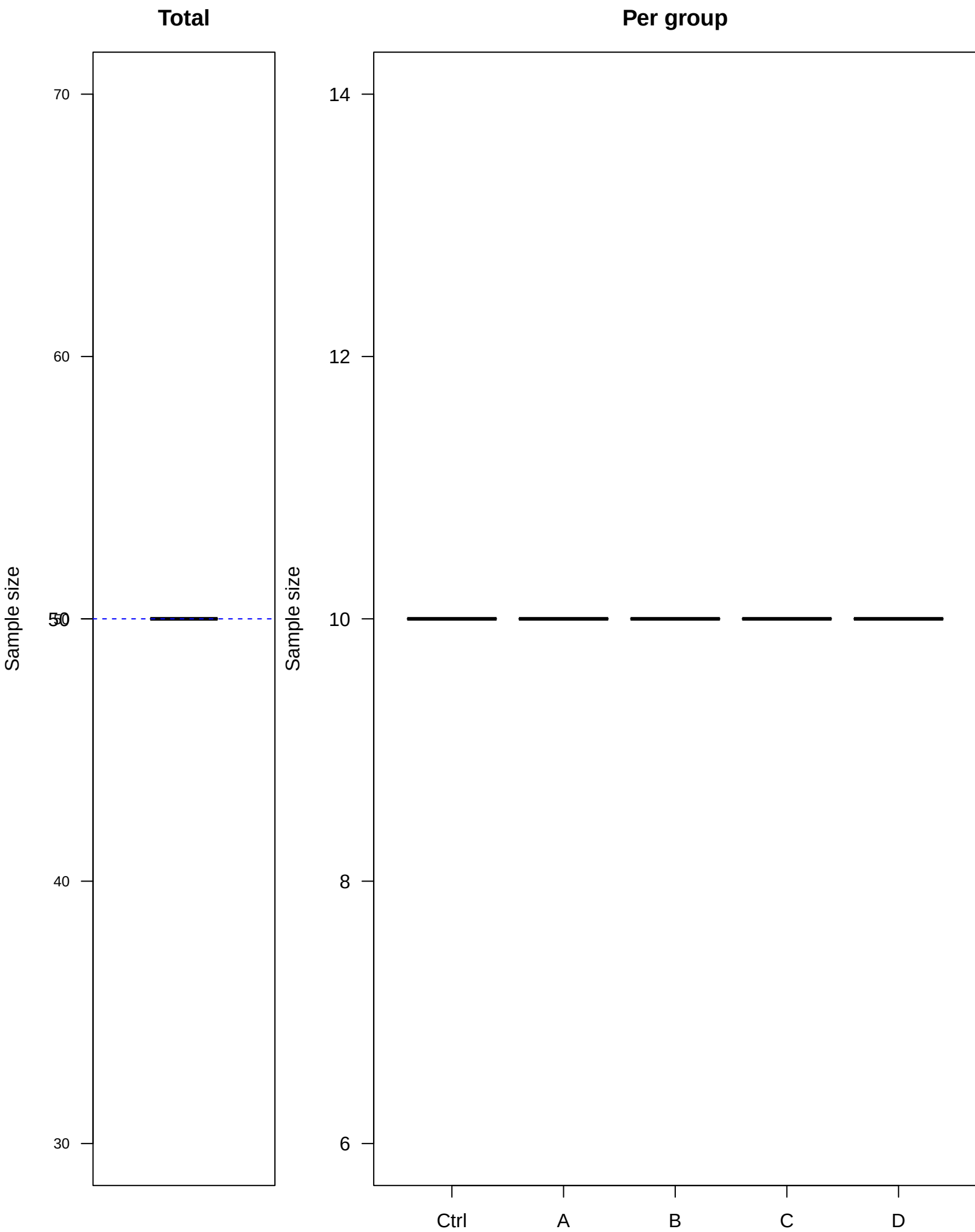
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 232 – sample size (H1) [binary]



Config 242 (0.05|0.9|4|1|0.95|0.5|0.05| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.950$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9643

FWER (P reject ≥ 1 | H0) : 0.0345

Max sample size (planned) : 25

ESS under H1 (overstated) : 25.0

ESS under H0 (overstated) : 25.0

NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 242 – sample size (H1) [binary]



Config 262 (0.05|0.9|4|2|0.52|0.5| 0.7| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.740 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9170

FWER (P reject ≥ 1 | H0) : 0.0525

Max sample size (planned) : 15930

ESS under H1 (overstated) : 15477.5

ESS under H0 (overstated) : 15923.4

NB: design powered for the binary test (delta/sd), so achieved

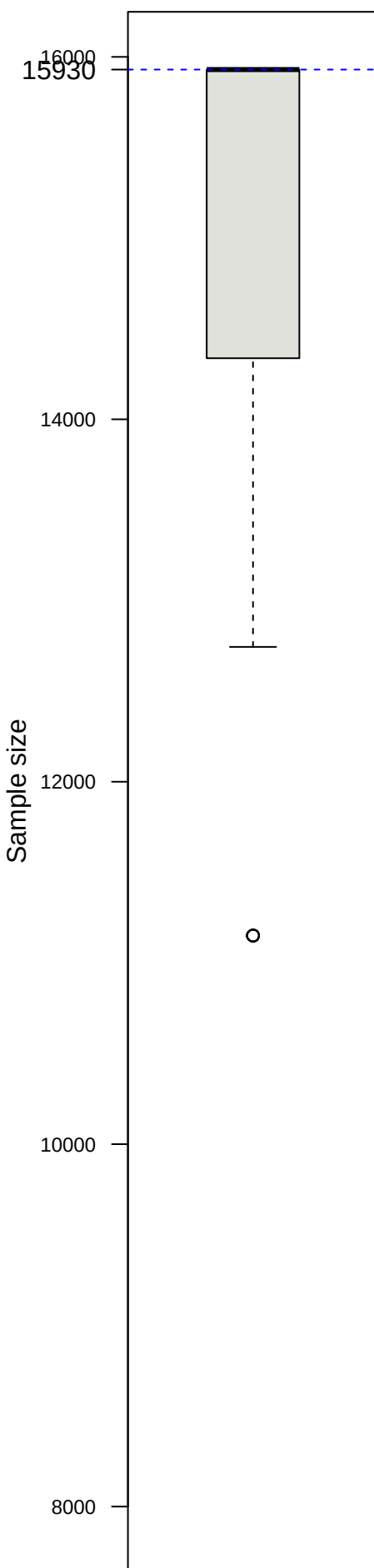
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

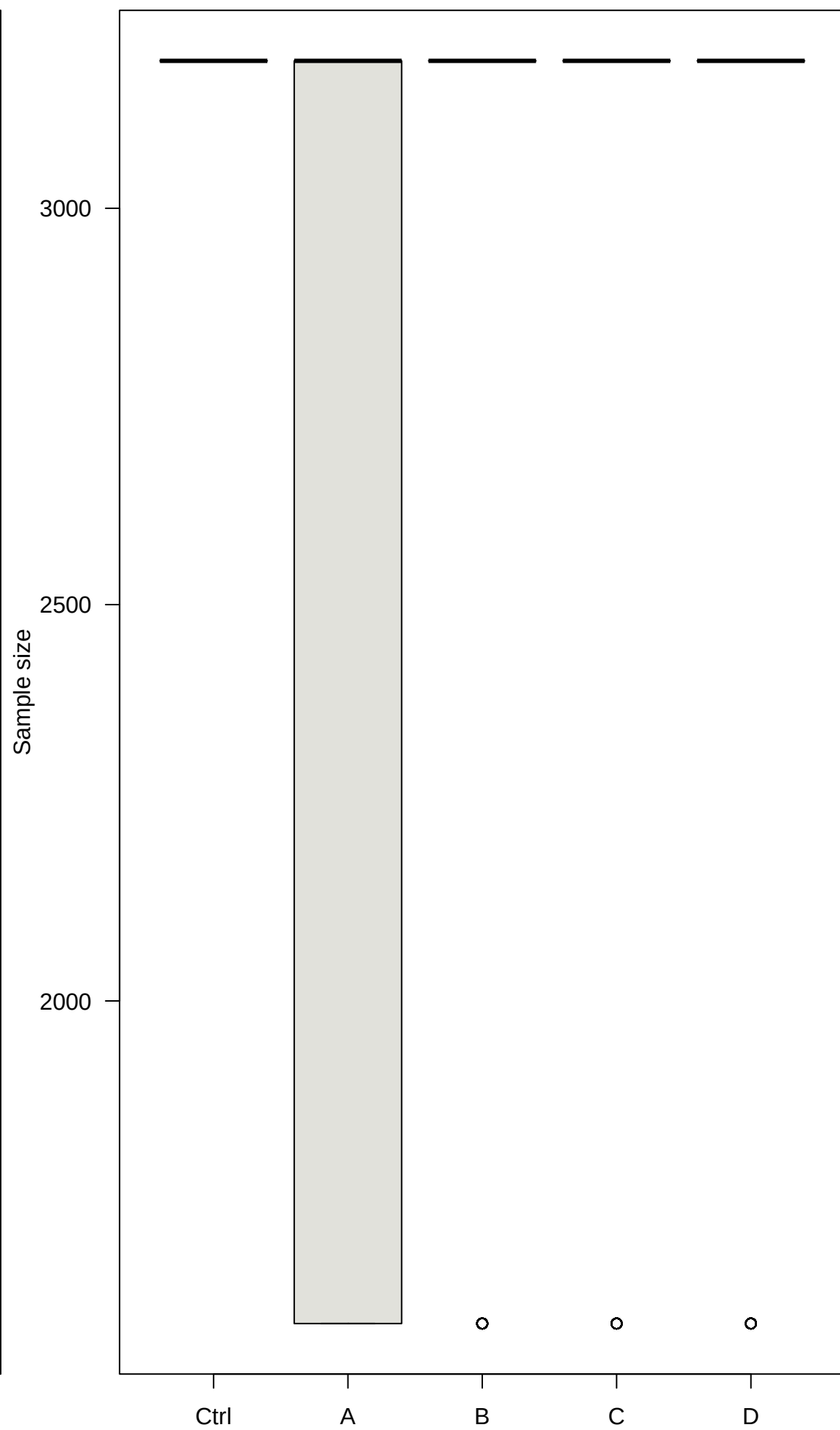
is overstated – see ../binary-binding for realistic ESS.

Config 262 – sample size (H1) [binary]

Total



Per group



Config 264 (0.05|0.9|4|2|0.52|0.5| 0.9| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.900

effect pi_t = 0.940 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8954

FWER (P reject ≥ 1 | H0) : 0.0496

Max sample size (planned) : 5810

ESS under H1 (overstated) : 5676.5

ESS under H0 (overstated) : 5807.6

NB: design powered for the binary test (delta/sd), so achieved

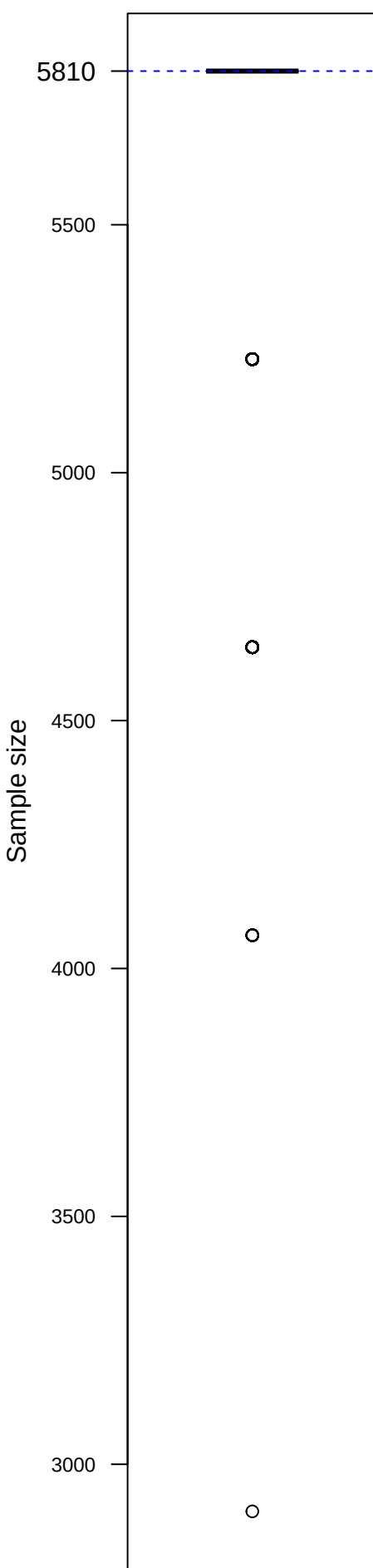
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

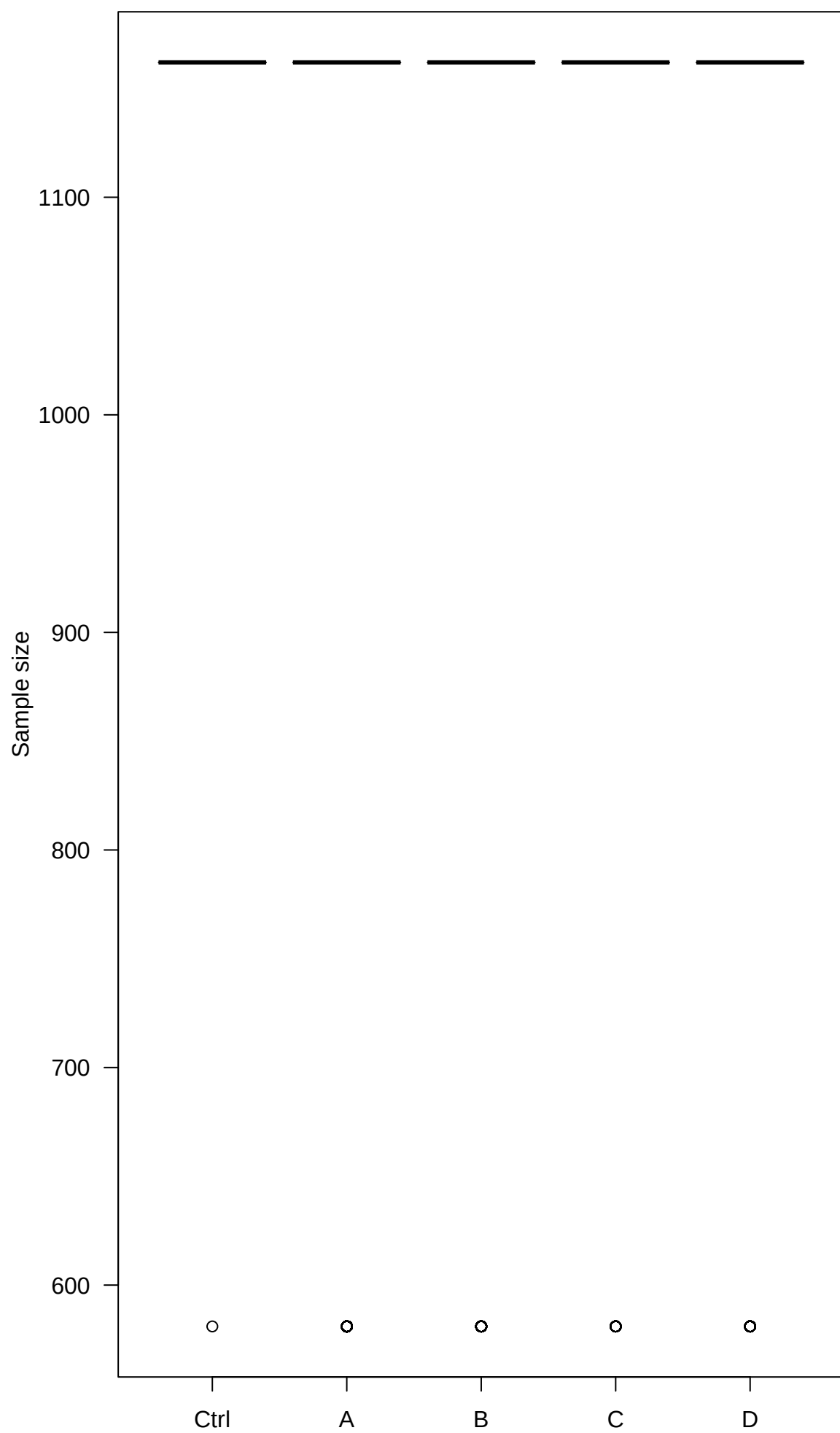
is overstated – see ../binary-binding for realistic ESS.

Config 264 – sample size (H1) [binary]

Total



Per group



Config 272 (0.05|0.9|4|2|0.55|0.5| 0.5| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9149

FWER (P reject ≥ 1 | H0) : 0.0492

Max sample size (planned) : 3120

ESS under H1 (overstated) : 3033.8

ESS under H0 (overstated) : 3118.8

NB: design powered for the binary test (delta/sd), so achieved

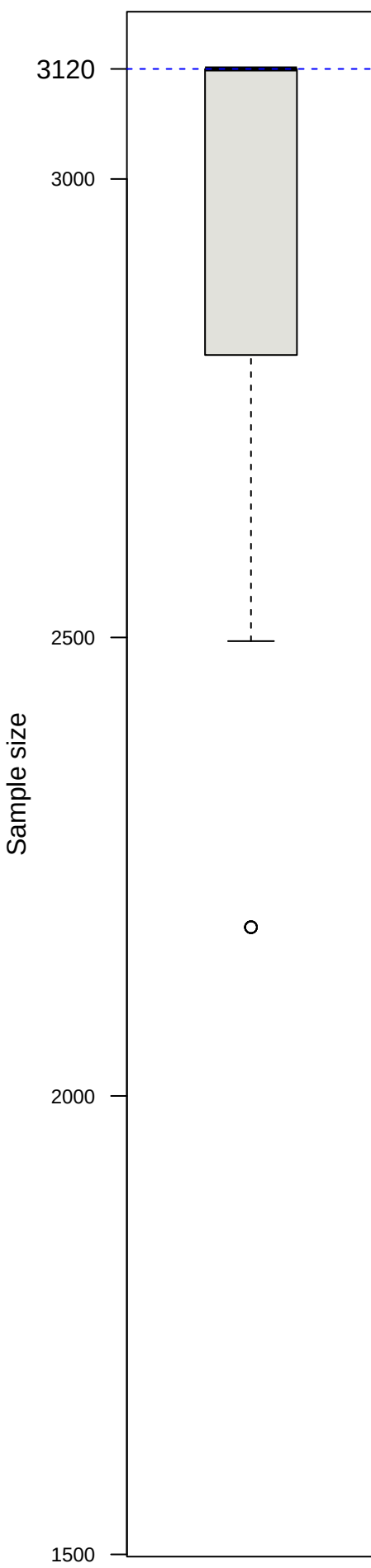
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

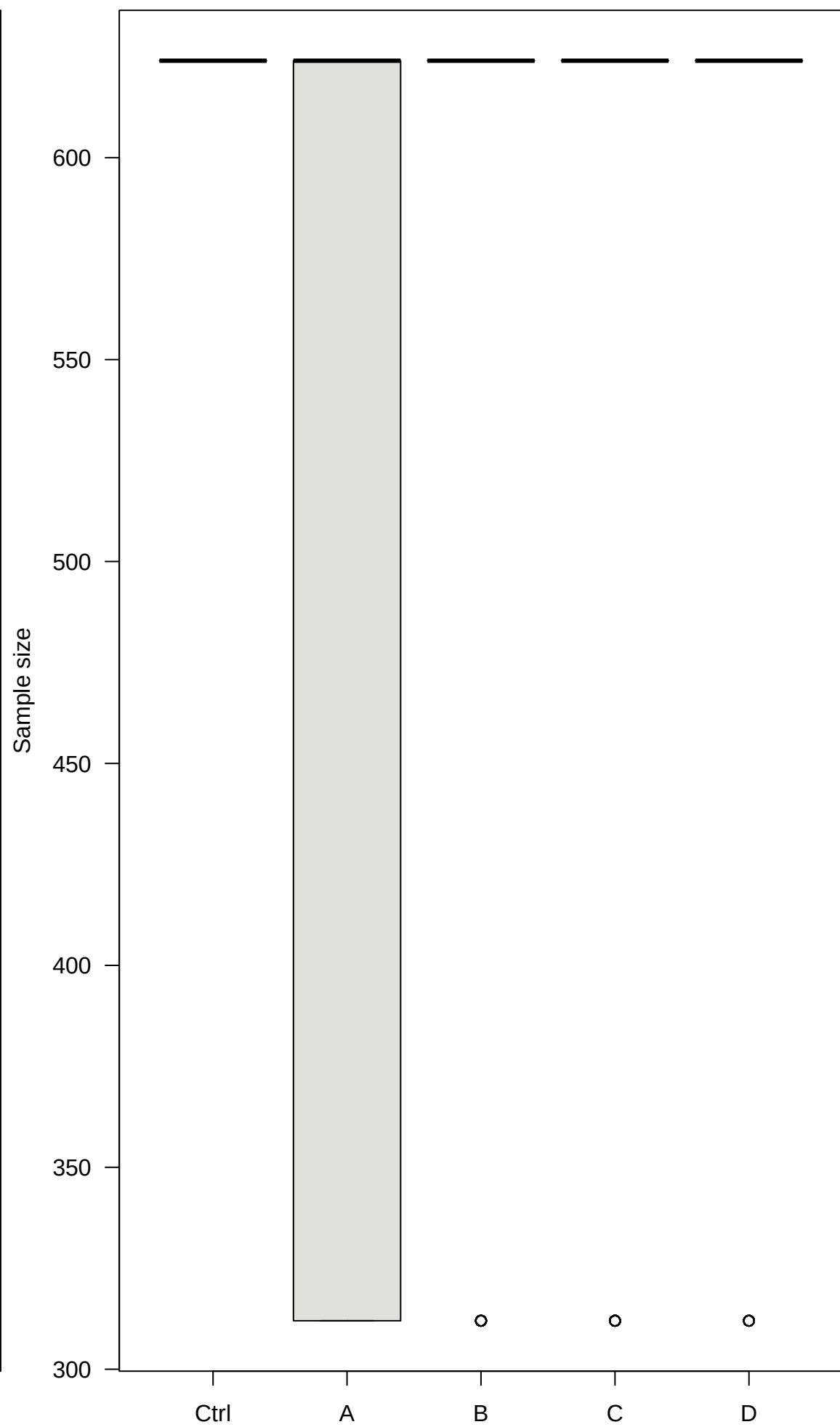
is overstated – see ../binary-binding for realistic ESS.

Config 272 – sample size (H1) [binary]

Total



Per group



Config 286 (0.05|0.9|4|2| 0.6|0.5| 0.7| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.900 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8875

FWER (P reject ≥ 1 | H0) : 0.0486

Max sample size (planned) : 500

ESS under H1 (overstated) : 490.4

ESS under H0 (overstated) : 499.8

NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 286 – sample size (H1) [binary]



Config 290 (0.05|0.9|4|2|0.65|0.5|0.05| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.050

effect pi_t = 0.350 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9830

FWER (P reject ≥ 1 | H0) : 0.0546

Max sample size (planned) : 220

ESS under H1 (overstated) : 207.6

ESS under H0 (overstated) : 219.9

NB: design powered for the binary test (delta/sd), so achieved

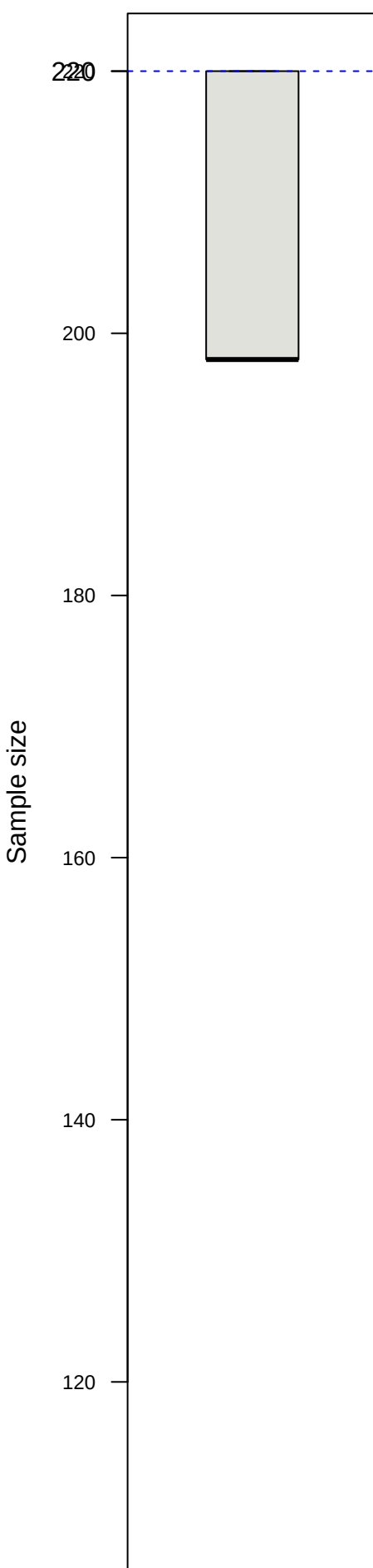
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

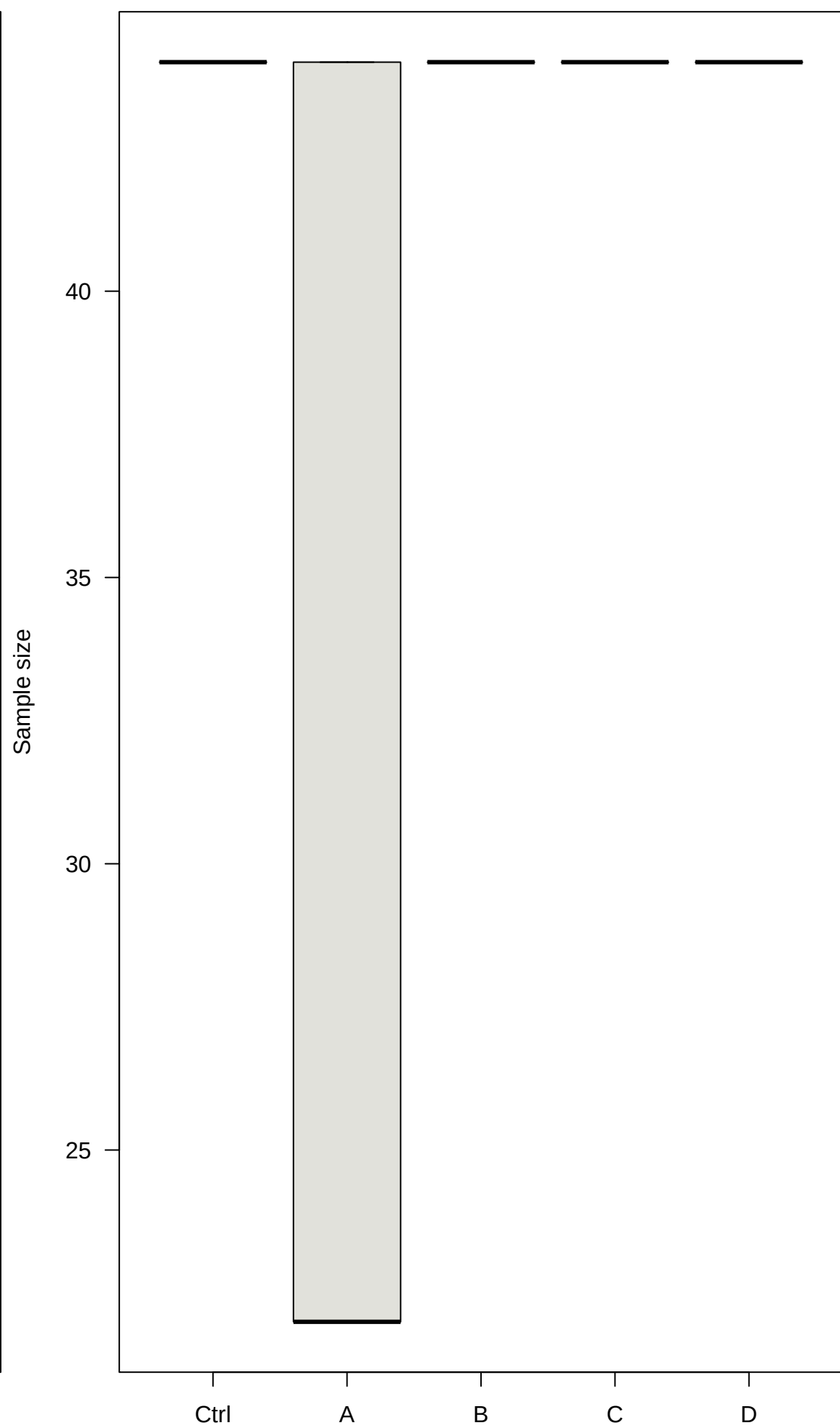
is overstated – see ../binary-binding for realistic ESS.

Config 290 – sample size (H1) [binary]

Total



Per group



Config 294 (0.05|0.9|4|2|0.65|0.5| 0.3| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9291

FWER (P reject ≥ 1 | H0) : 0.0524

Max sample size (planned) : 340

ESS under H1 (overstated) : 329.6

ESS under H0 (overstated) : 339.9

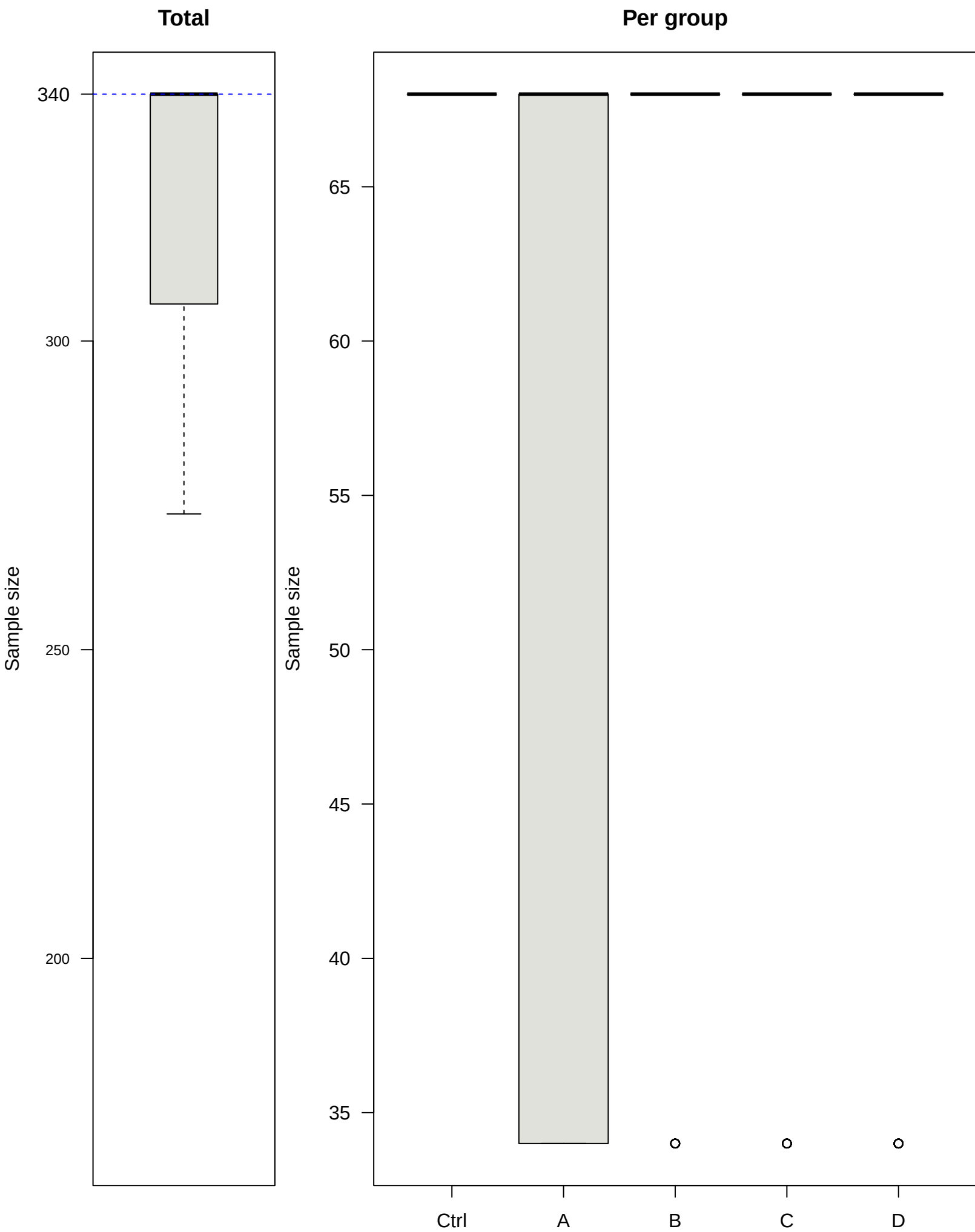
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 294 – sample size (H1) [binary]



Config 296 (0.05|0.9|4|2|0.65|0.5| 0.5| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9036

FWER (P reject ≥ 1 | H0) : 0.0416

Max sample size (planned) : 310

ESS under H1 (overstated) : 303.1

ESS under H0 (overstated) : 309.9

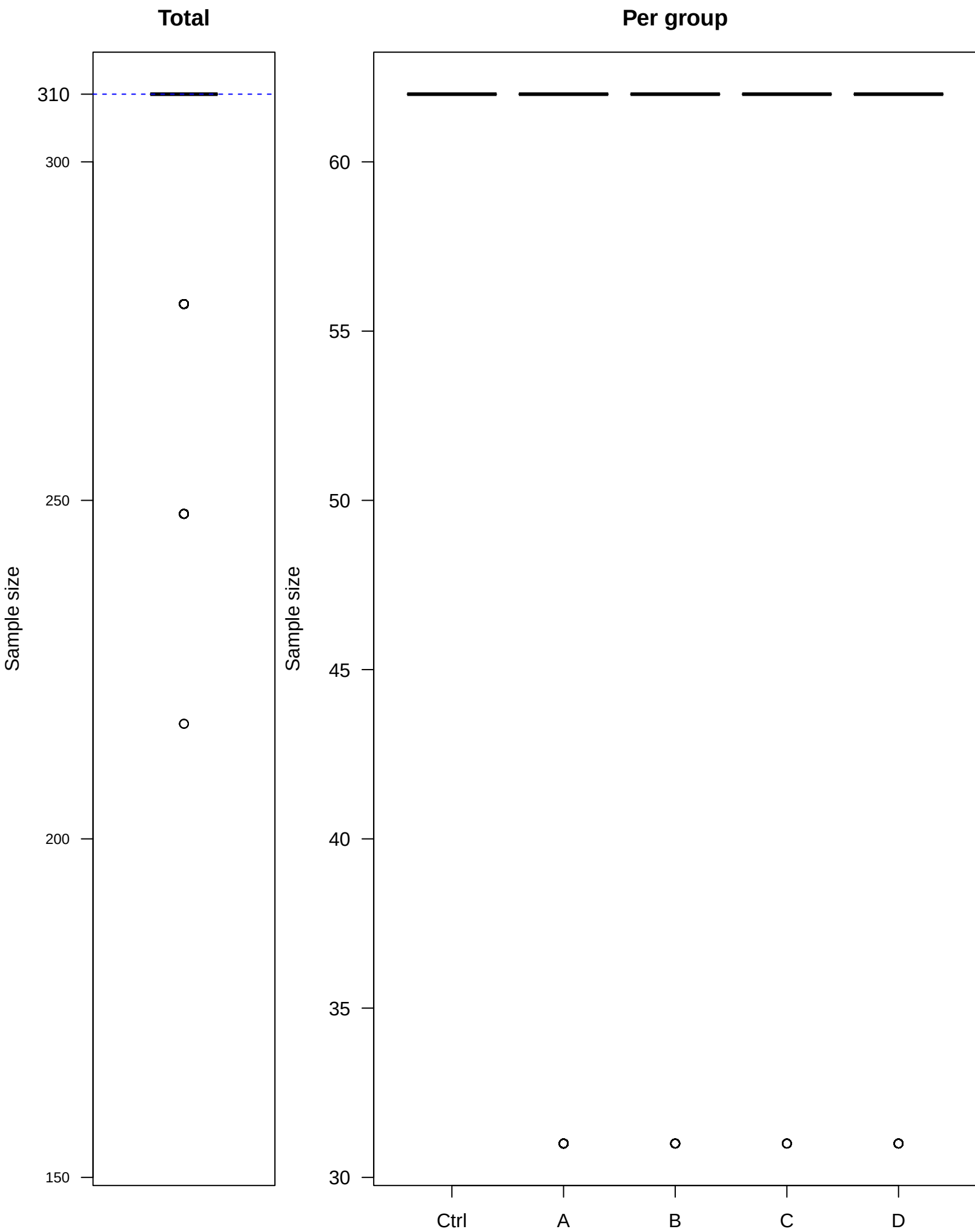
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 296 – sample size (H1) [binary]



Config 306 (0.05|0.9|4|2|0.75|0.5| 0.3| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9315

FWER (P reject ≥ 1 | H0) : 0.0518

Max sample size (planned) : 120

ESS under H1 (overstated) : 117.0

ESS under H0 (overstated) : 120.0

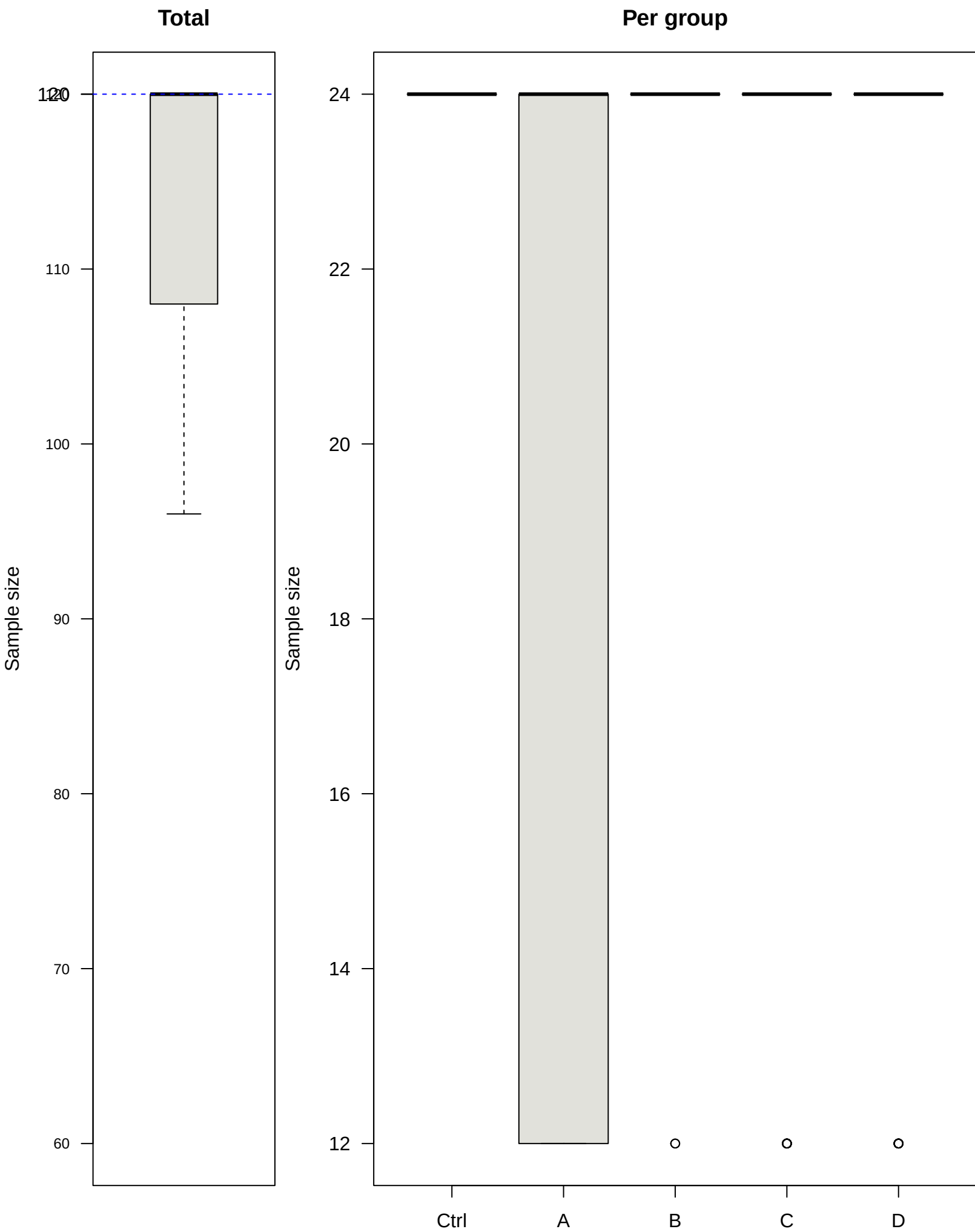
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 306 – sample size (H1) [binary]



Config 316 (0.05|0.9|4|2|0.85|0.5| 0.1| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.100

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9538

FWER (P reject ≥ 1 | H0) : 0.0658

Max sample size (planned) : 50

ESS under H1 (overstated) : 48.4

ESS under H0 (overstated) : 50.0

NB: design powered for the binary test (delta/sd), so achieved

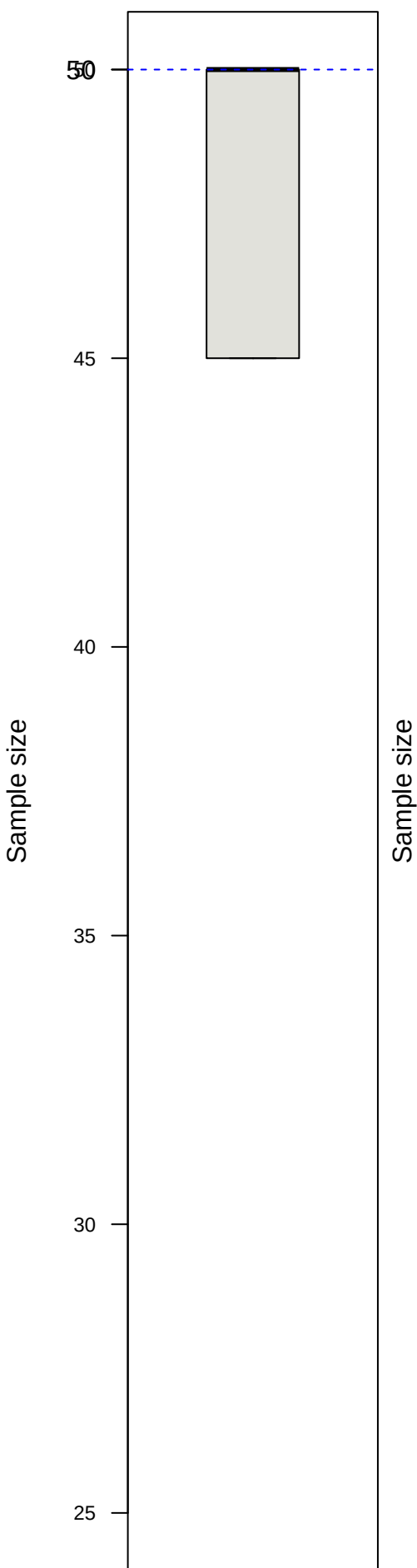
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

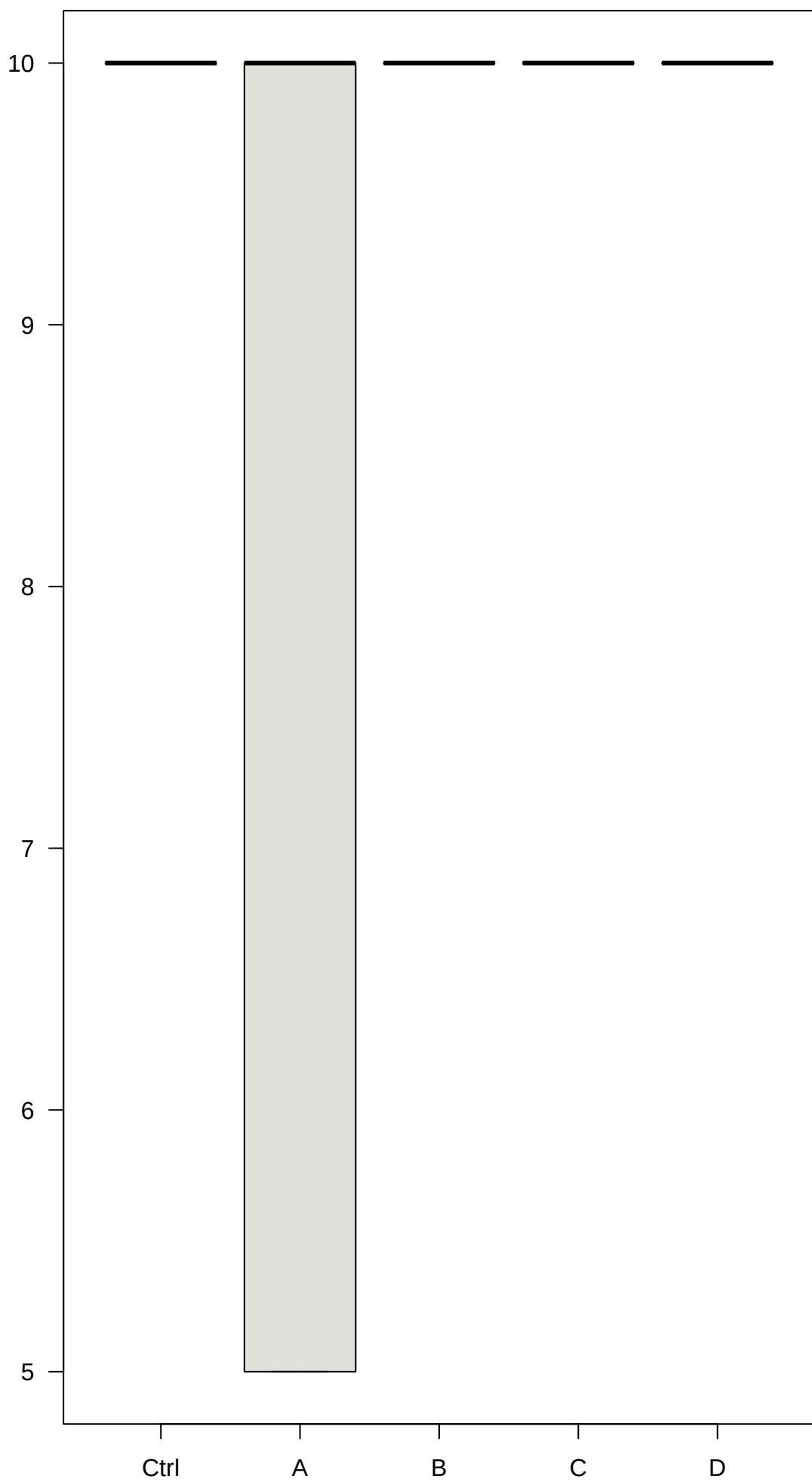
is overstated – see ../binary-binding for realistic ESS.

Config 316 – sample size (H1) [binary]

Total



Per group



Config 326 (0.05|0.9|4|2|0.95|0.5|0.05| sep) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.050

effect pi_t = 0.950 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9842

FWER (P reject ≥ 1 | H0) : 0.0420

Max sample size (planned) : 30

ESS under H1 (overstated) : 30.0

ESS under H0 (overstated) : 30.0

NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 326 – sample size (H1) [binary]

